

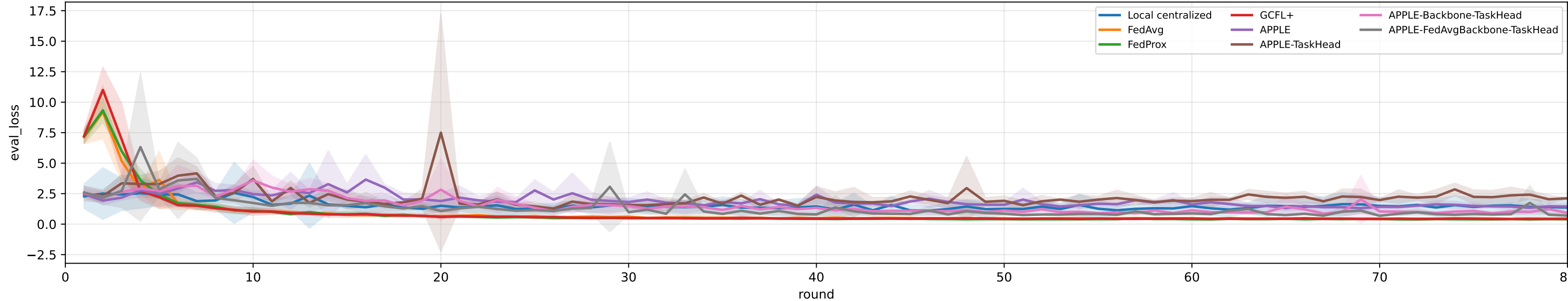
Controlled q-label heterogeneity: Mid | aggregated / pooled performance over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

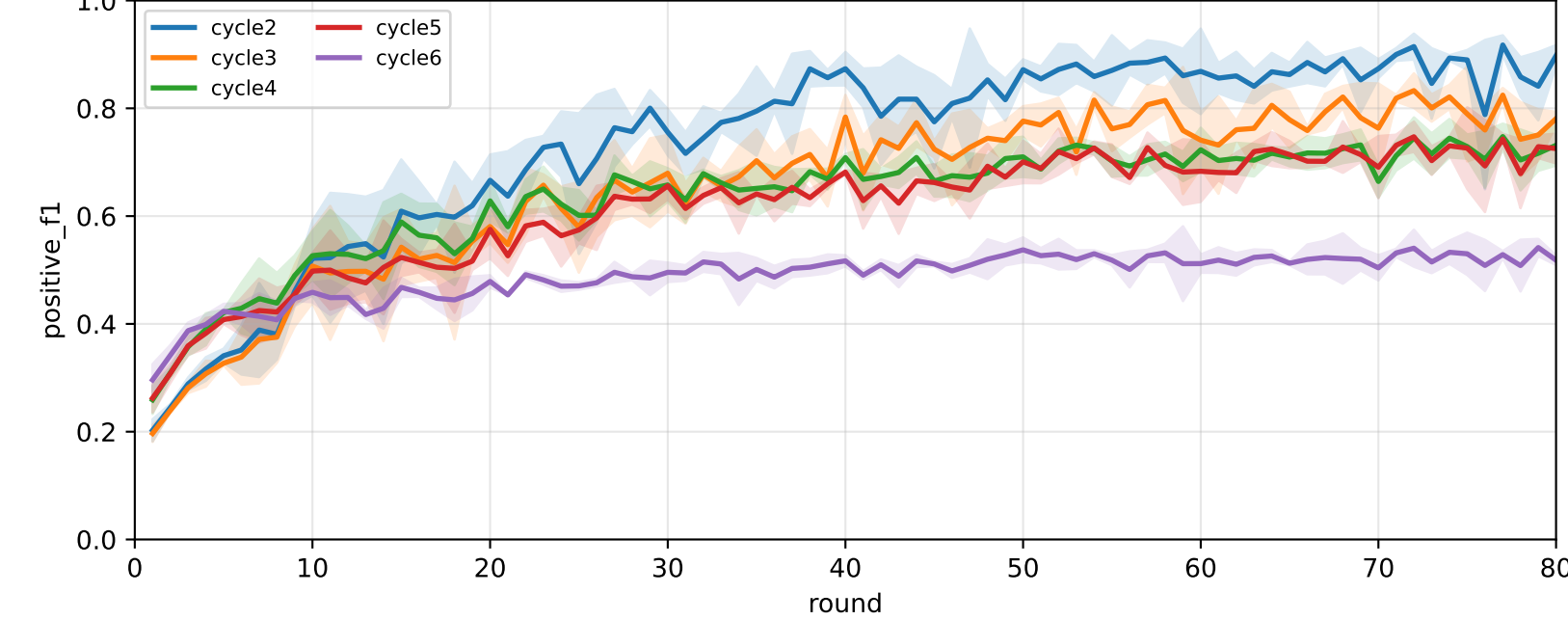
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	60.3% ± 0.4%	63.4% ± 0.4%	74.7% ± 4.2%	70.7% ± 0.1%	60.9% ± 2.0%	57.0% ± 1.1%	47.0% ± 0.4%
FedAvg	63.8% ± 1.5%	66.4% ± 1.6%	80.3% ± 4.9%	74.7% ± 5.1%	63.2% ± 1.5%	60.7% ± 0.8%	53.0% ± 2.5%
FedProx	66.4% ± 1.2%	68.8% ± 0.9%	84.0% ± 1.6%	74.1% ± 4.4%	66.5% ± 0.9%	64.2% ± 0.5%	55.2% ± 2.4%
GCFL+	66.2% ± 4.2%	68.9% ± 3.6%	82.9% ± 3.6%	76.6% ± 7.4%	68.0% ± 3.8%	63.0% ± 5.1%	54.0% ± 4.4%
APPLE	47.3% ± 5.1%	47.4% ± 5.3%	43.7% ± 8.8%	47.9% ± 4.5%	50.1% ± 7.2%	46.2% ± 3.8%	48.8% ± 2.4%
APPLE-TaskHead	51.9% ± 4.6%	53.0% ± 5.3%	56.3% ± 12.6%	55.1% ± 2.6%	51.1% ± 4.5%	50.4% ± 5.1%	47.1% ± 2.9%
APPLE-Backbone-TaskHead	66.1% ± 2.8%	68.1% ± 3.1%	78.0% ± 3.0%	76.2% ± 7.6%	64.9% ± 2.3%	65.3% ± 2.5%	54.8% ± 1.6%
APPLE-FedAvgBackbone-TaskHead	69.1% ± 1.6%	71.8% ± 1.0%	84.5% ± 4.7%	81.0% ± 4.3%	67.7% ± 1.0%	67.7% ± 2.5%	56.9% ± 3.8%

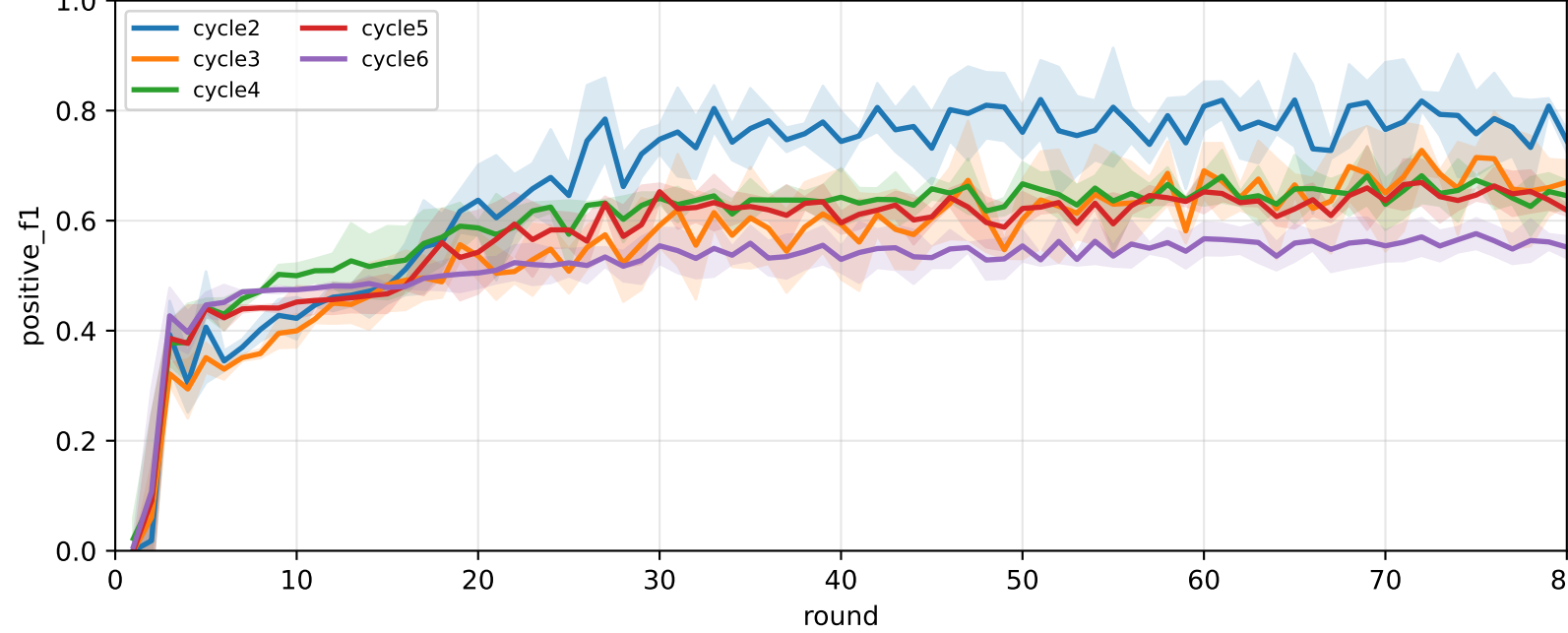
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



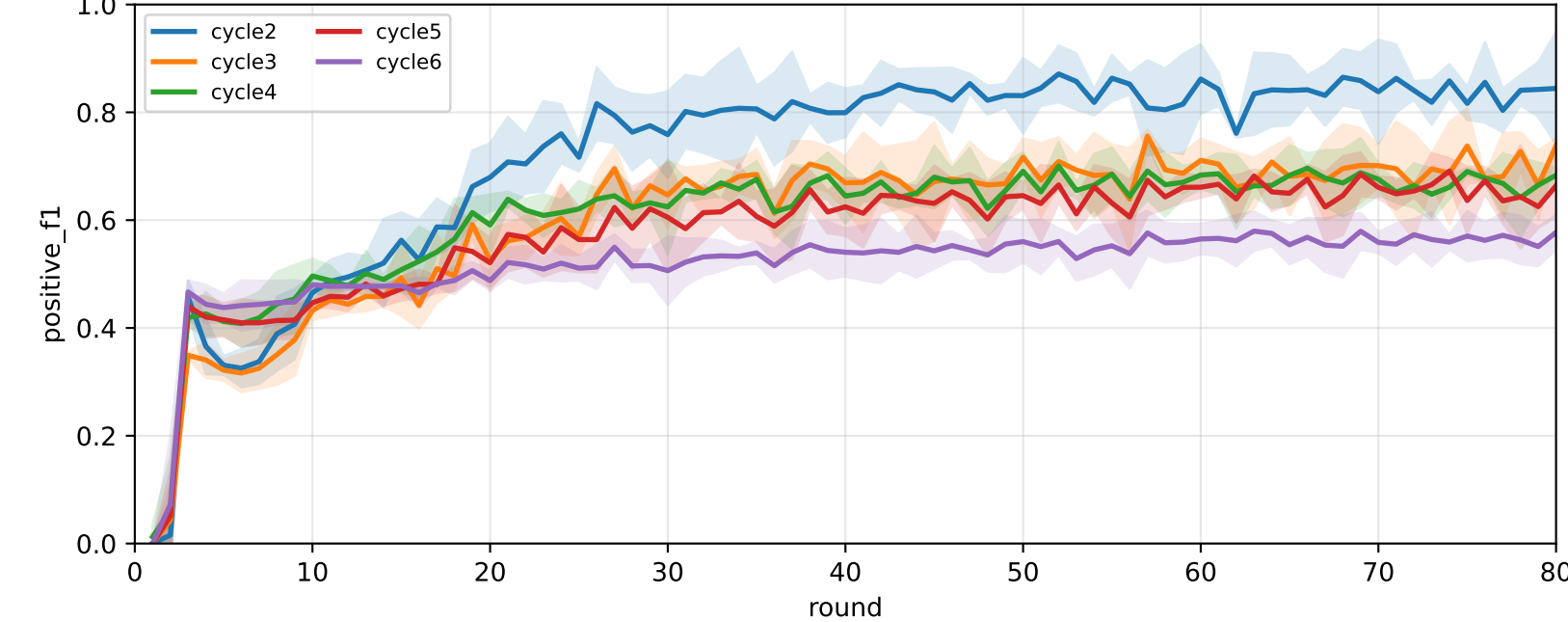
Pooled local centralized positive F1



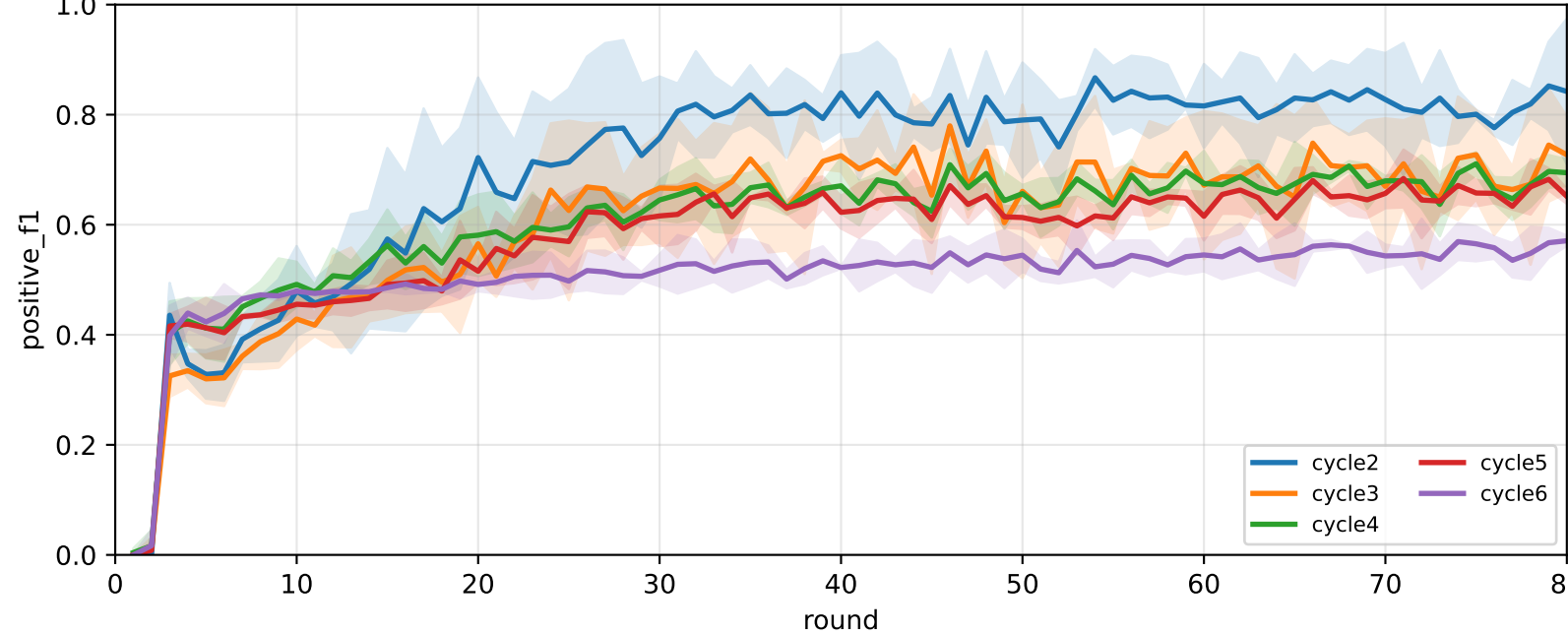
Pooled FedAvg positive F1



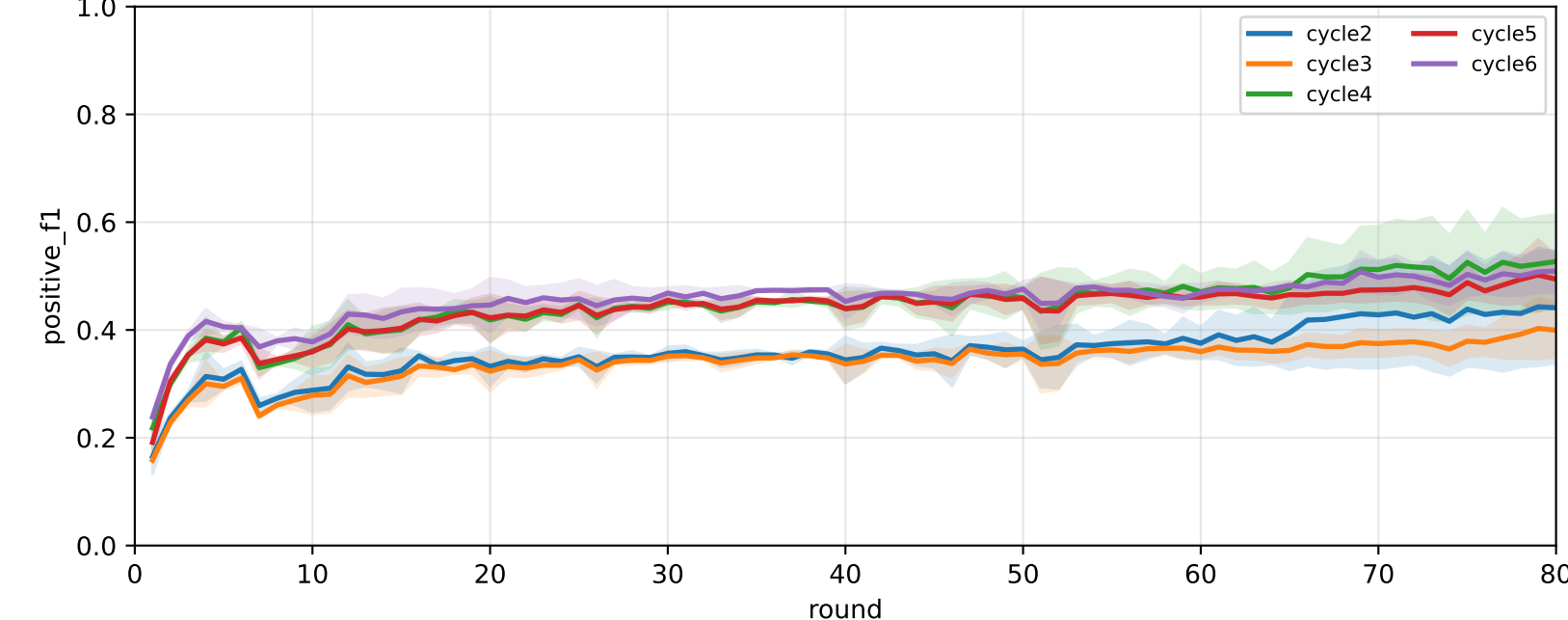
Pooled FedProx positive F1



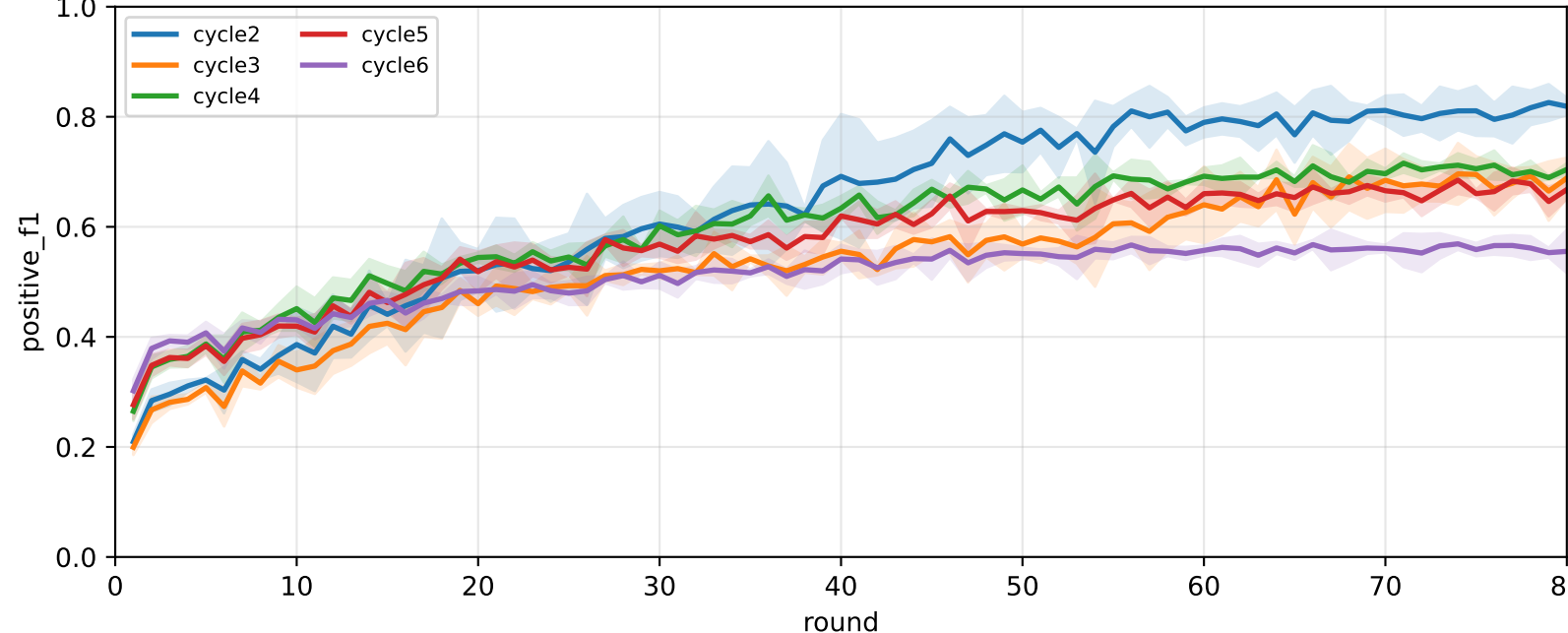
Pooled GCFL+ positive F1



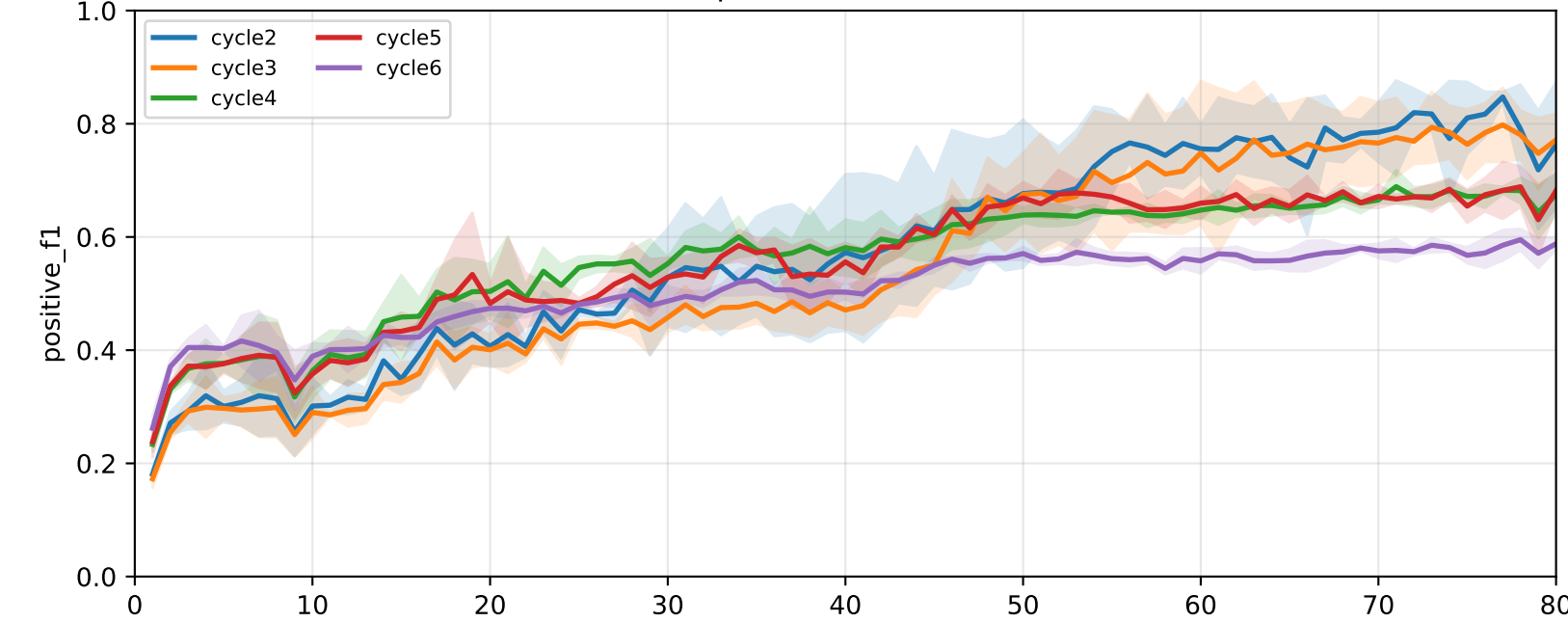
Pooled APPLE positive F1 (after-local selected model, rounds=80)



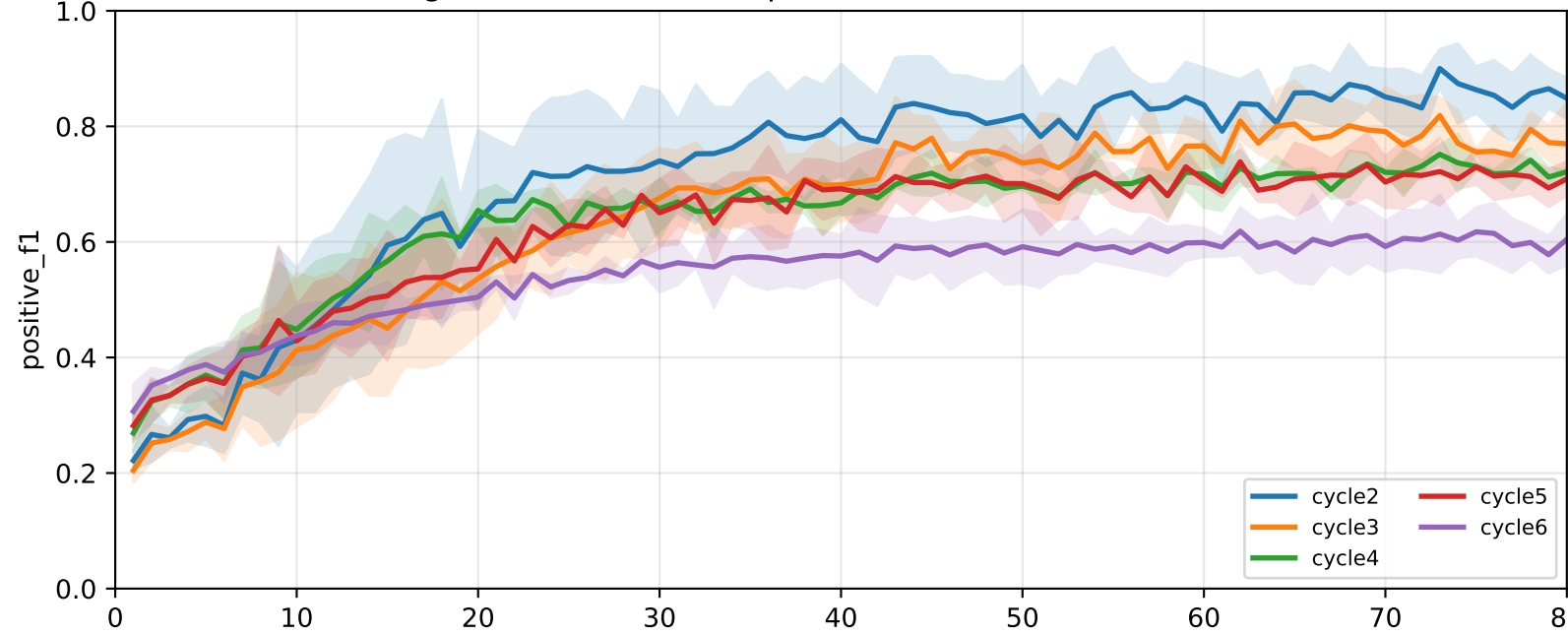
Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)



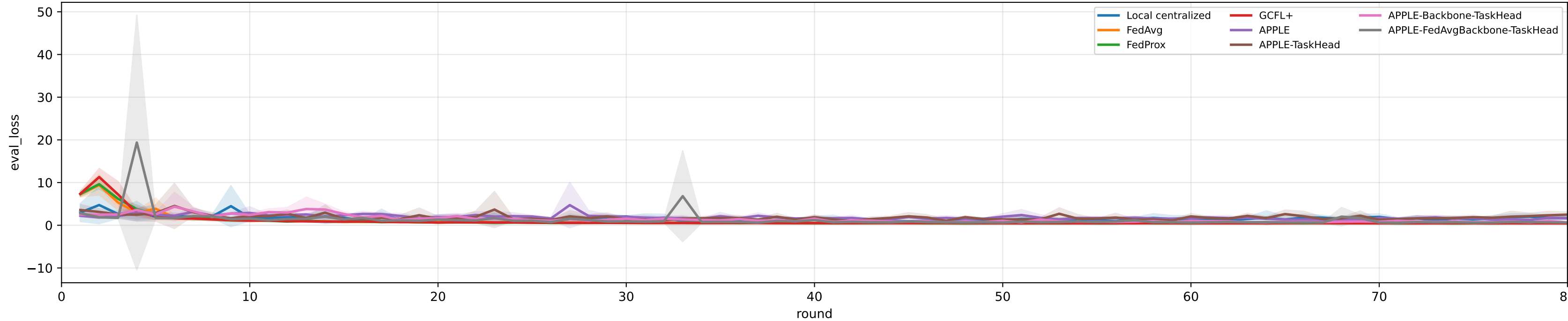
Controlled q-label heterogeneity: Mid | family aggregated / pooled performance |
family=q_task_cycle2 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

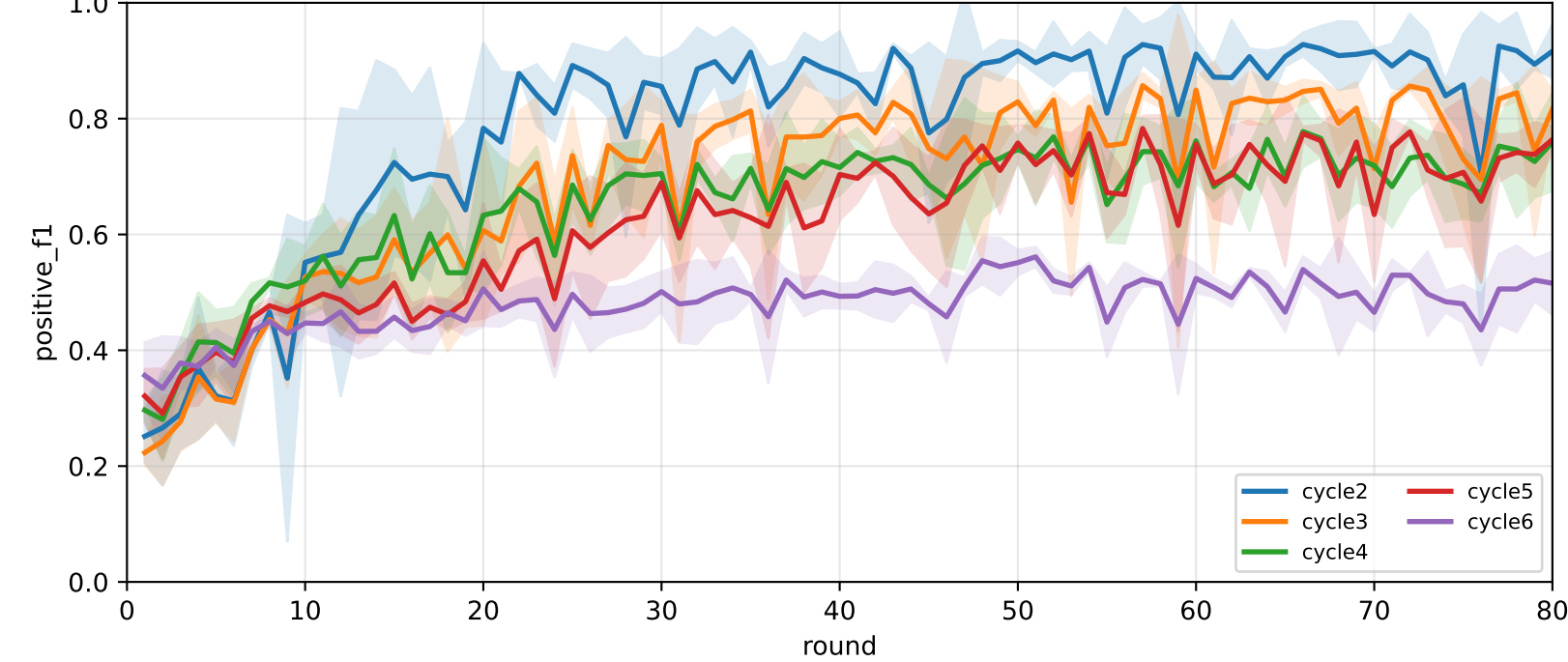
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	62.9% ± 2.7%	67.2% ± 2.6%	84.2% ± 3.3%	78.8% ± 3.0%	67.1% ± 2.5%	59.7% ± 2.9%	46.4% ± 2.7%
FedAvg	64.2% ± 1.6%	66.8% ± 1.7%	81.2% ± 5.6%	75.1% ± 5.0%	63.3% ± 1.7%	61.3% ± 1.0%	53.2% ± 2.6%
FedProx	66.5% ± 1.1%	69.0% ± 0.9%	85.2% ± 0.5%	73.8% ± 4.6%	66.4% ± 1.2%	64.4% ± 0.3%	55.4% ± 2.4%
GCFL+	67.0% ± 4.4%	69.7% ± 3.9%	83.5% ± 4.3%	77.4% ± 7.5%	68.8% ± 4.2%	64.3% ± 6.2%	54.4% ± 4.1%
APPLE	48.0% ± 6.0%	48.0% ± 6.2%	44.3% ± 9.6%	49.2% ± 6.1%	50.5% ± 8.6%	46.7% ± 4.4%	49.4% ± 2.7%
APPLE-TaskHead	54.1% ± 2.3%	54.4% ± 2.6%	53.3% ± 8.1%	57.0% ± 3.4%	53.7% ± 4.1%	55.3% ± 1.2%	52.8% ± 2.4%
APPLE-Backbone-TaskHead	63.6% ± 3.4%	66.2% ± 3.4%	81.3% ± 4.3%	72.0% ± 8.9%	62.4% ± 1.1%	62.7% ± 5.2%	52.5% ± 1.9%
APPLE-FedAvgBackbone-TaskHead	70.2% ± 1.0%	72.9% ± 0.9%	83.0% ± 9.1%	85.3% ± 3.3%	71.4% ± 4.3%	66.6% ± 4.0%	58.1% ± 4.4%

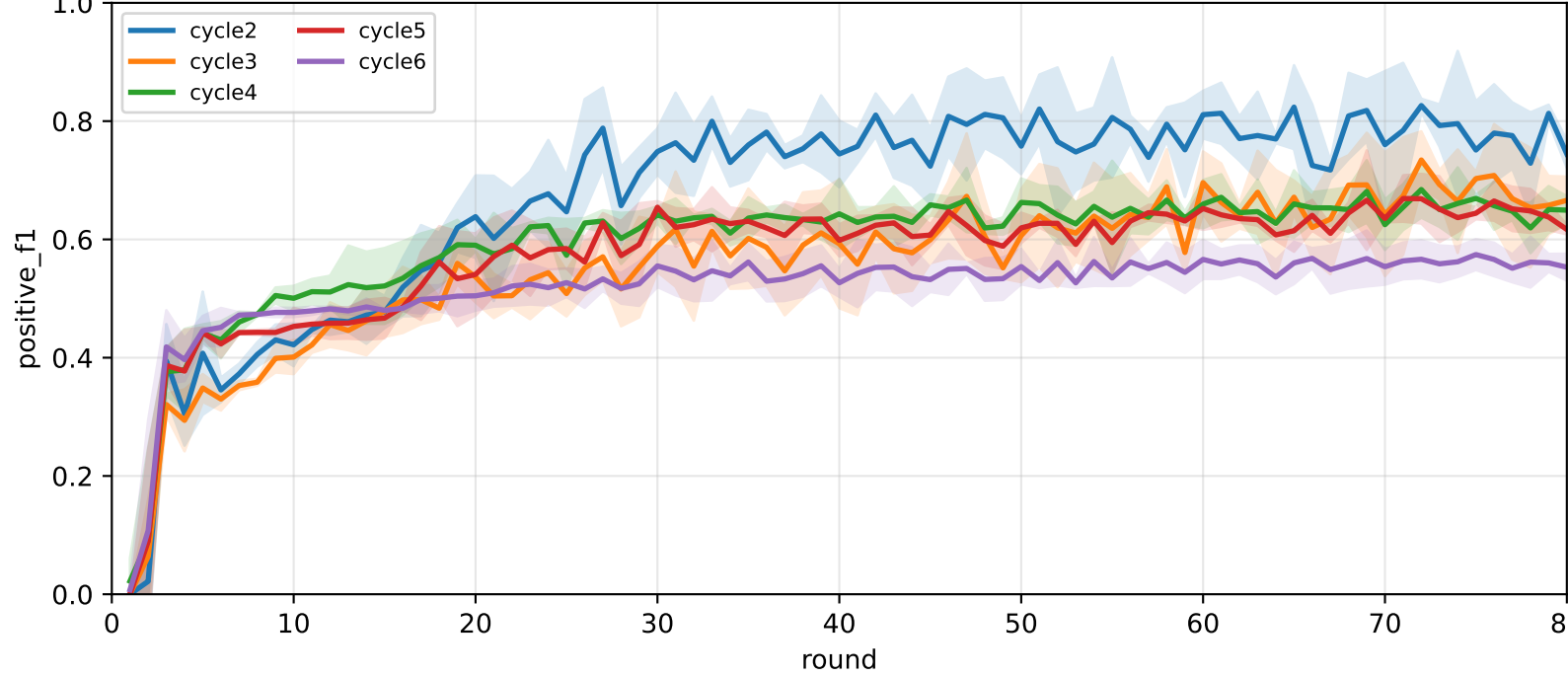
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



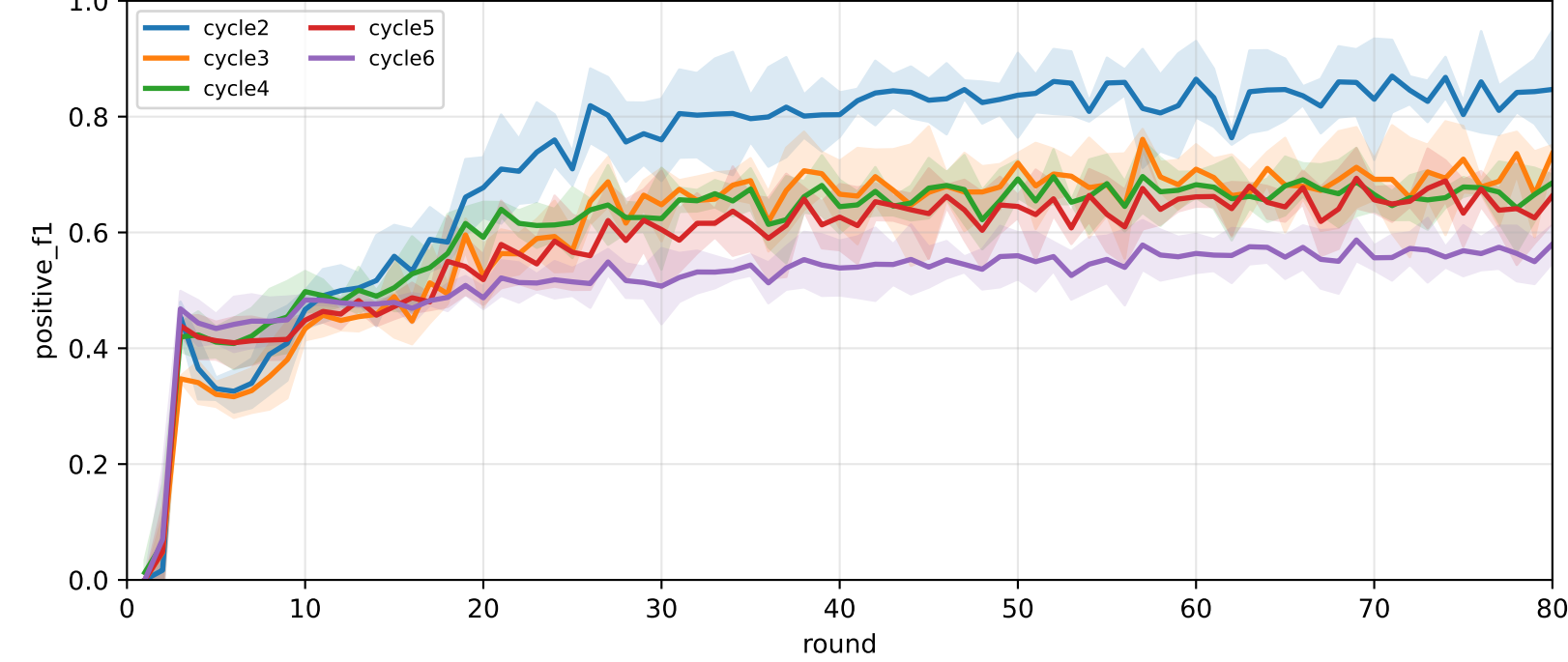
Pooled local centralized positive F1



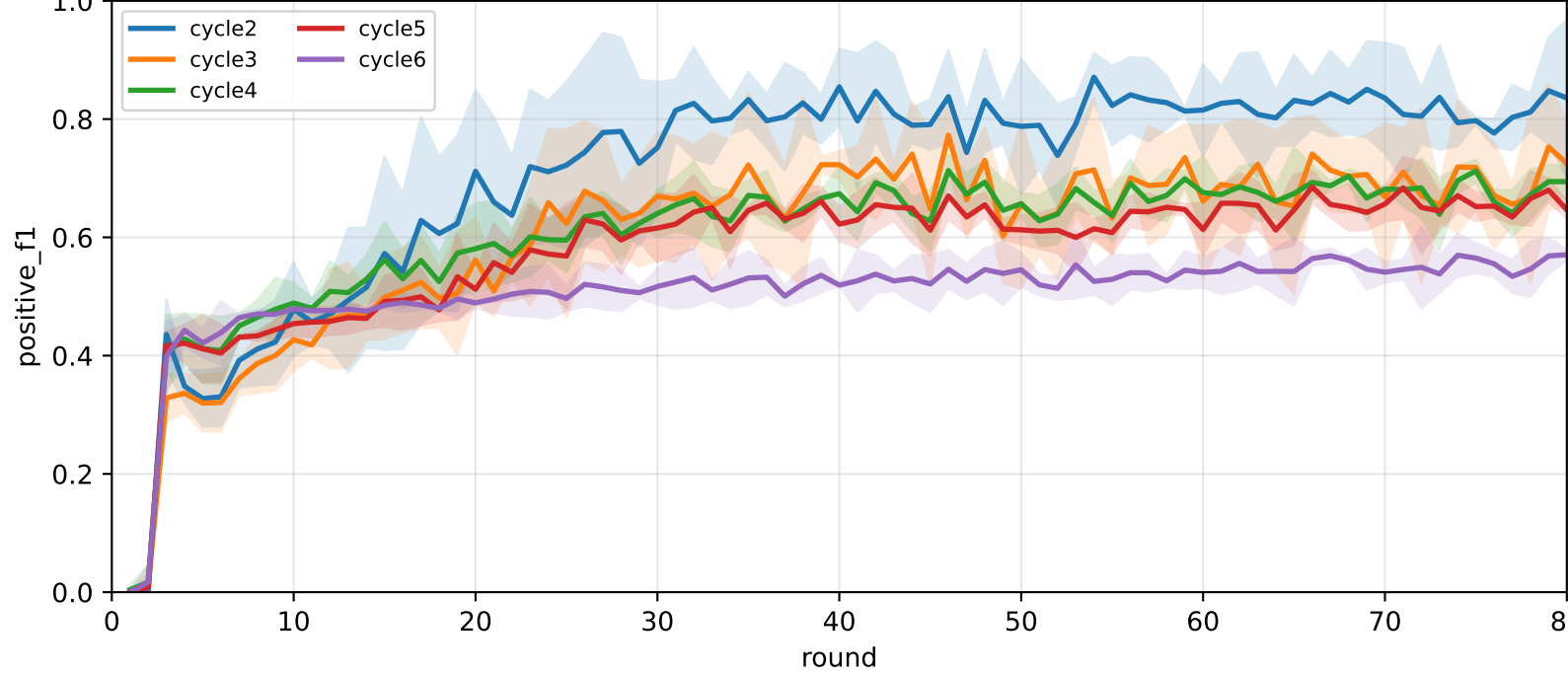
Pooled FedAvg positive F1



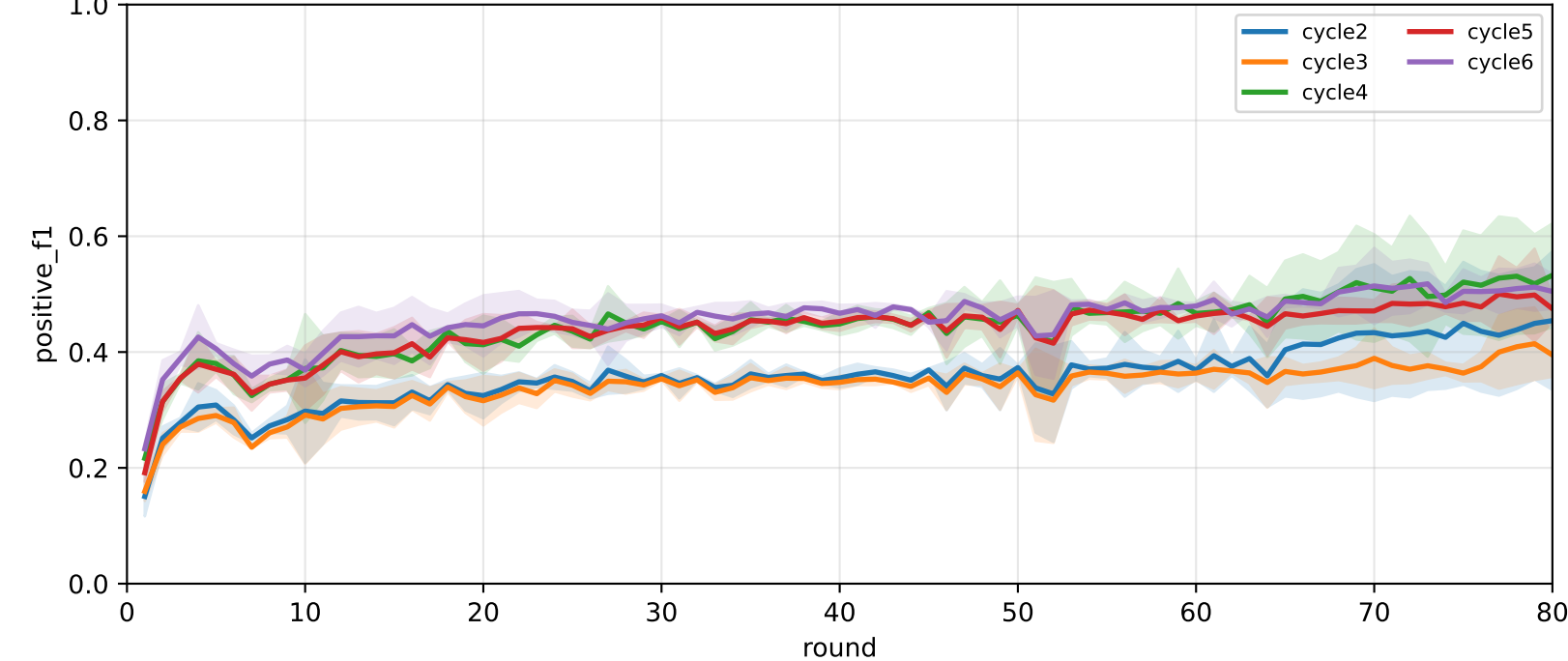
Pooled FedProx positive F1



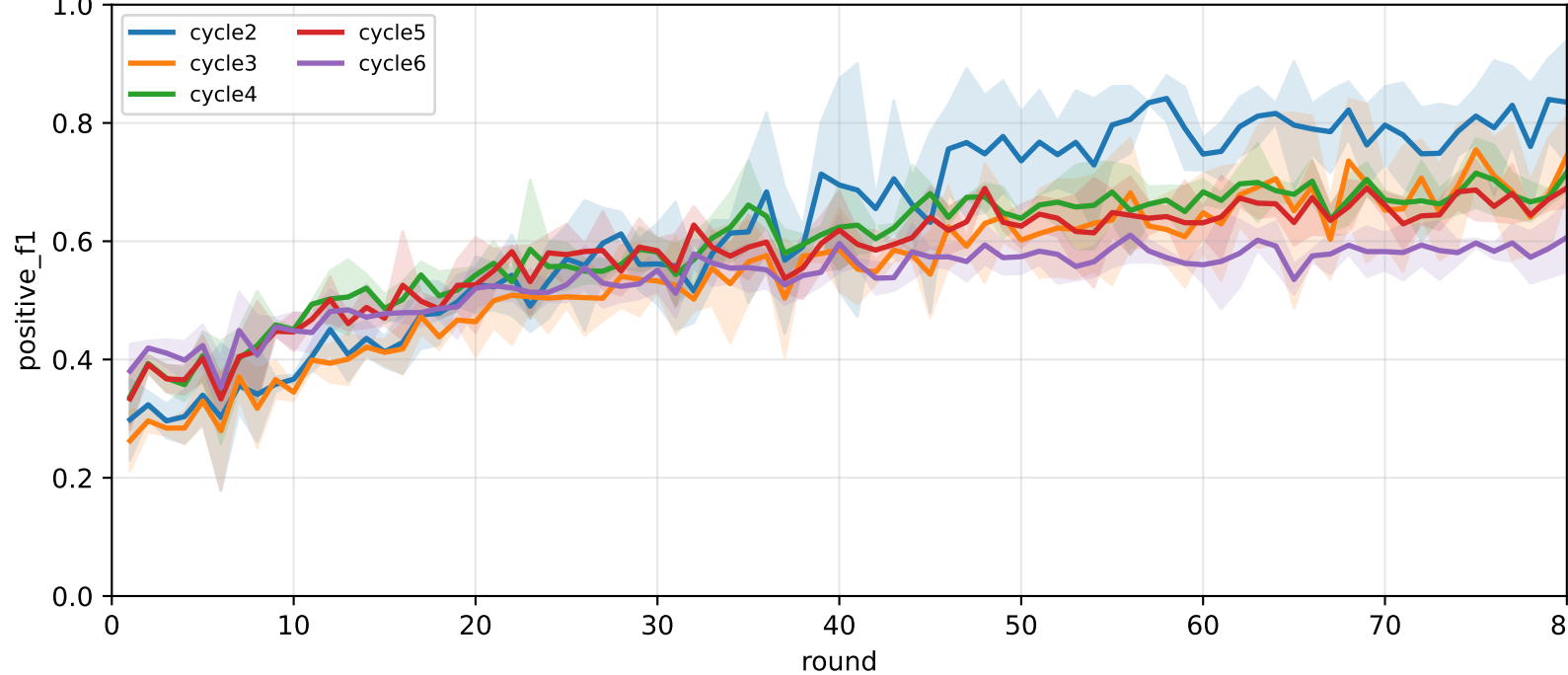
Pooled GCFL+ positive F1



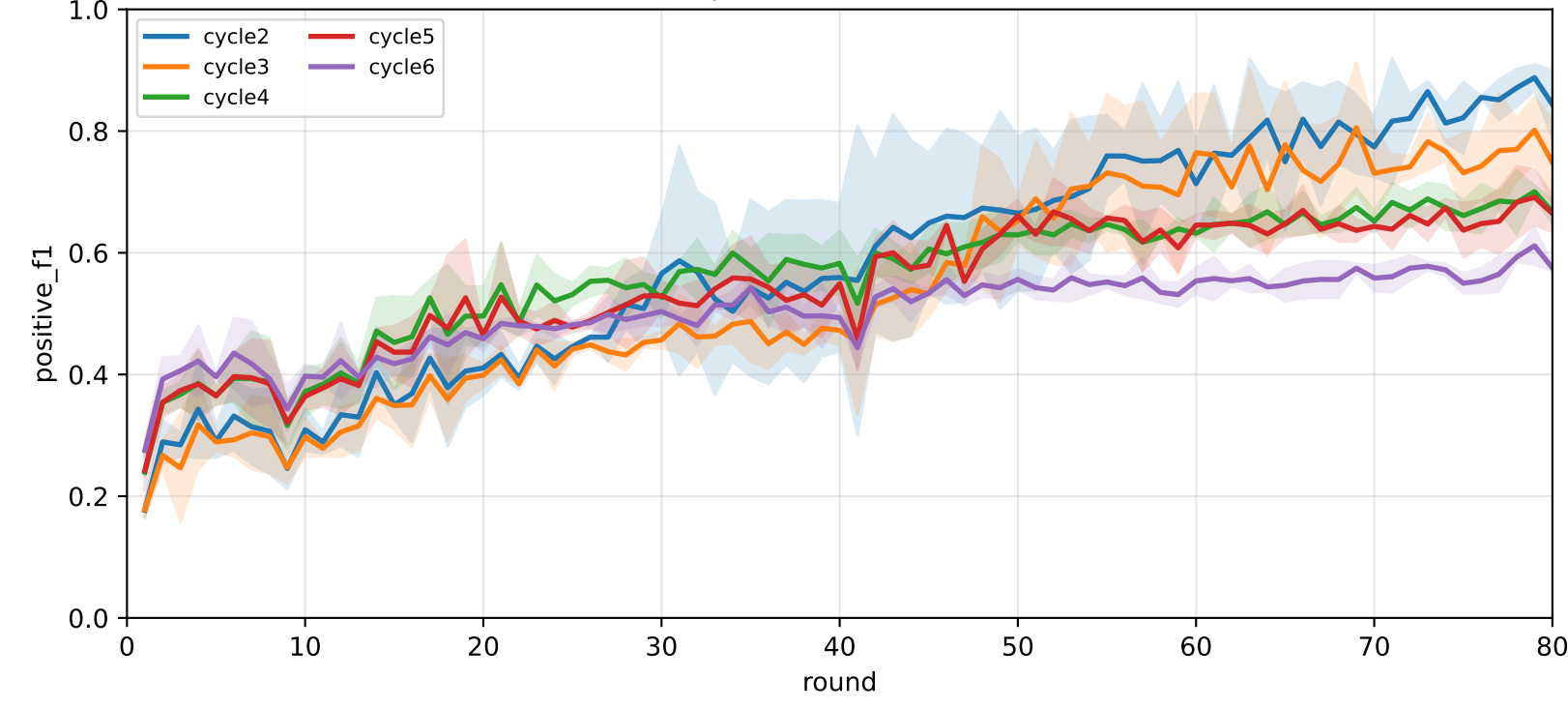
Pooled APPLE positive F1 (after-local selected model, rounds=80)



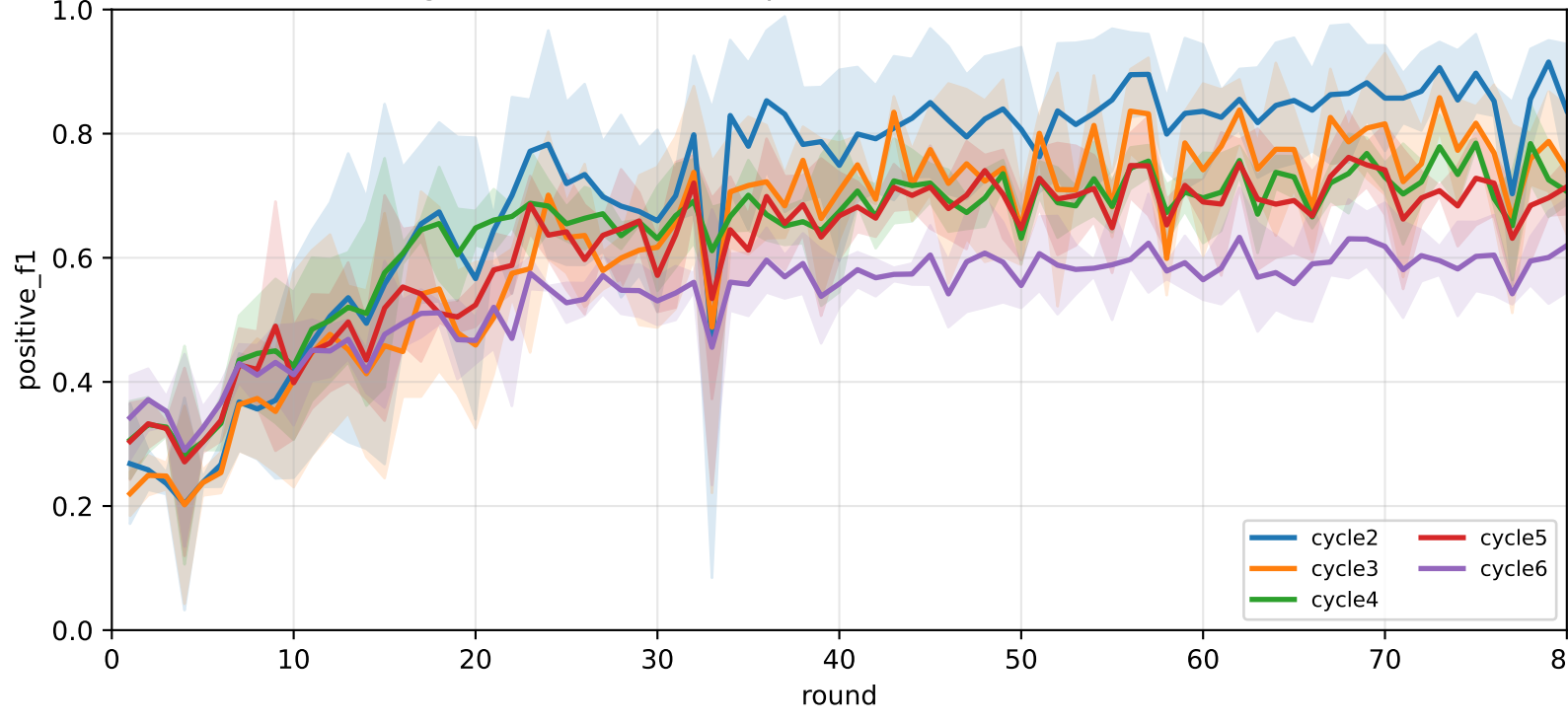
Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)



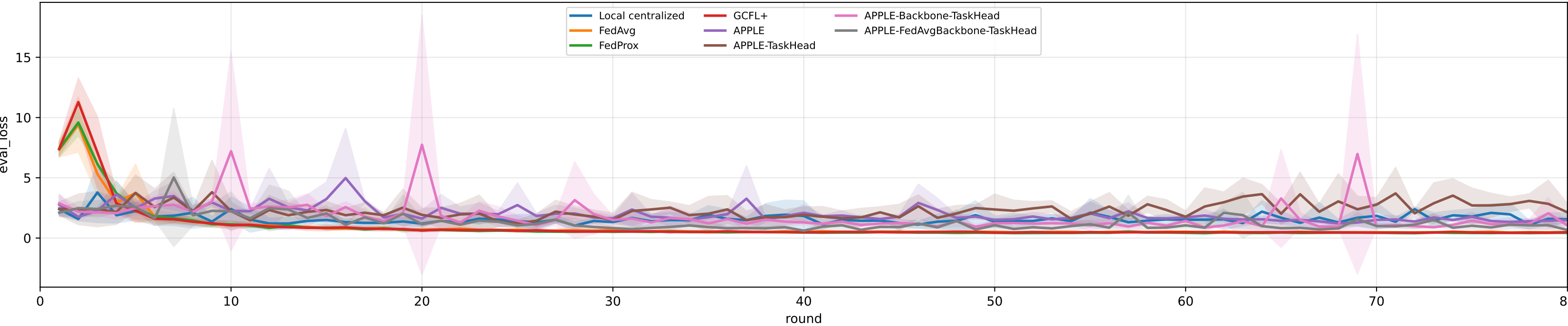
Controlled q-label heterogeneity: Mid | family aggregated / pooled performance |
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model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

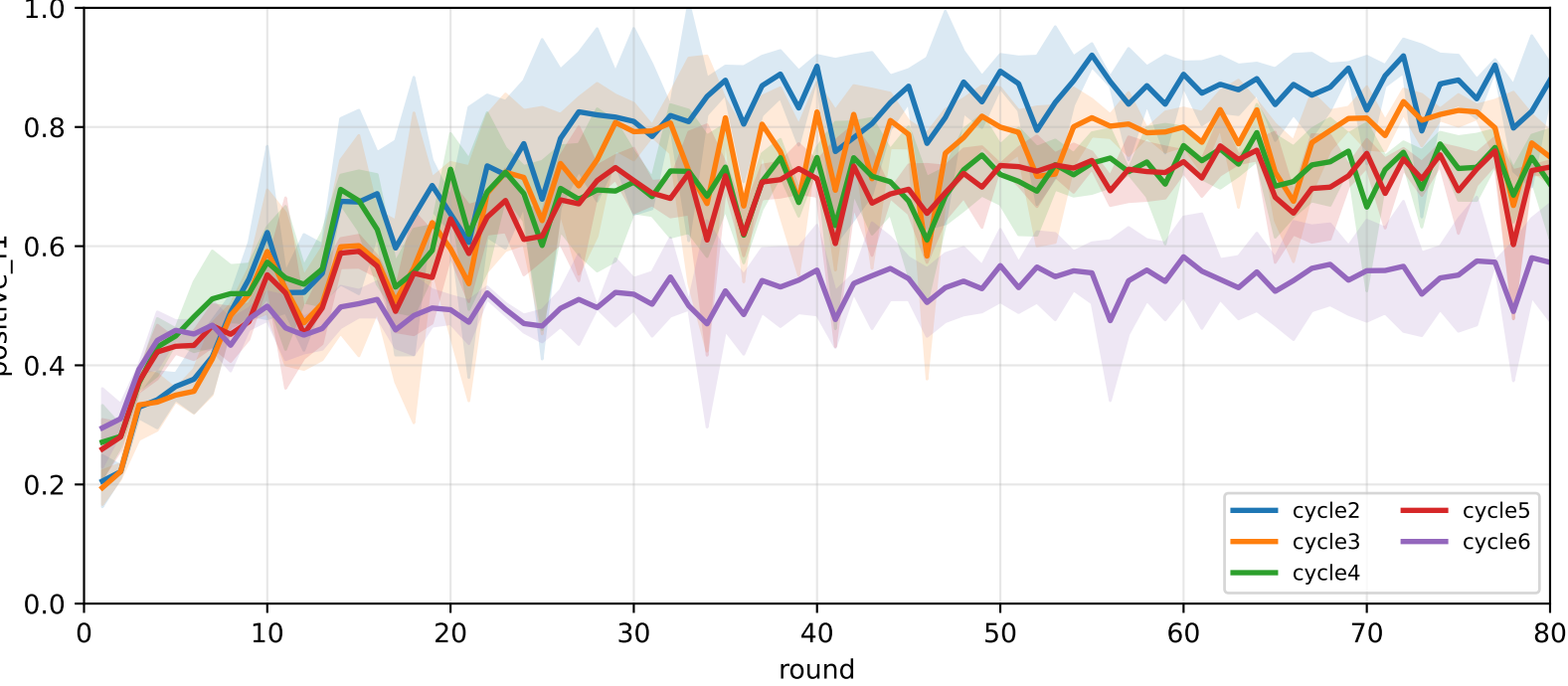
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	55.7% ± 8.8%	57.9% ± 10.2%	66.6% ± 17.9%	62.6% ± 13.2%	58.0% ± 7.4%	56.9% ± 12.7%	45.4% ± 3.0%
FedAvg	63.6% ± 1.9%	66.1% ± 1.7%	80.1% ± 4.6%	74.4% ± 5.9%	63.1% ± 1.7%	60.1% ± 1.2%	52.8% ± 2.9%
FedProx	66.5% ± 1.6%	69.0% ± 1.1%	84.3% ± 2.5%	74.9% ± 5.4%	66.7% ± 1.3%	63.9% ± 0.9%	55.3% ± 2.8%
GCFL+	65.9% ± 4.0%	68.7% ± 3.4%	82.4% ± 3.2%	77.0% ± 7.1%	67.5% ± 4.0%	62.7% ± 4.7%	53.7% ± 4.4%
APPLE	46.9% ± 4.7%	47.1% ± 5.1%	44.3% ± 9.9%	46.7% ± 3.7%	50.8% ± 8.1%	45.2% ± 2.8%	48.4% ± 2.1%
APPLE-TaskHead	50.3% ± 9.4%	51.8% ± 10.0%	58.9% ± 19.3%	55.0% ± 5.0%	53.5% ± 7.4%	48.2% ± 10.9%	43.2% ± 8.8%
APPLE-Backbone-TaskHead	68.6% ± 2.5%	70.4% ± 2.8%	77.8% ± 3.9%	78.8% ± 6.3%	71.5% ± 2.9%	68.4% ± 0.6%	55.5% ± 2.0%
APPLE-FedAvgBackbone-TaskHead	69.4% ± 4.0%	71.8% ± 3.2%	85.8% ± 5.4%	78.2% ± 10.8%	67.6% ± 1.9%	68.9% ± 4.4%	58.5% ± 5.4%

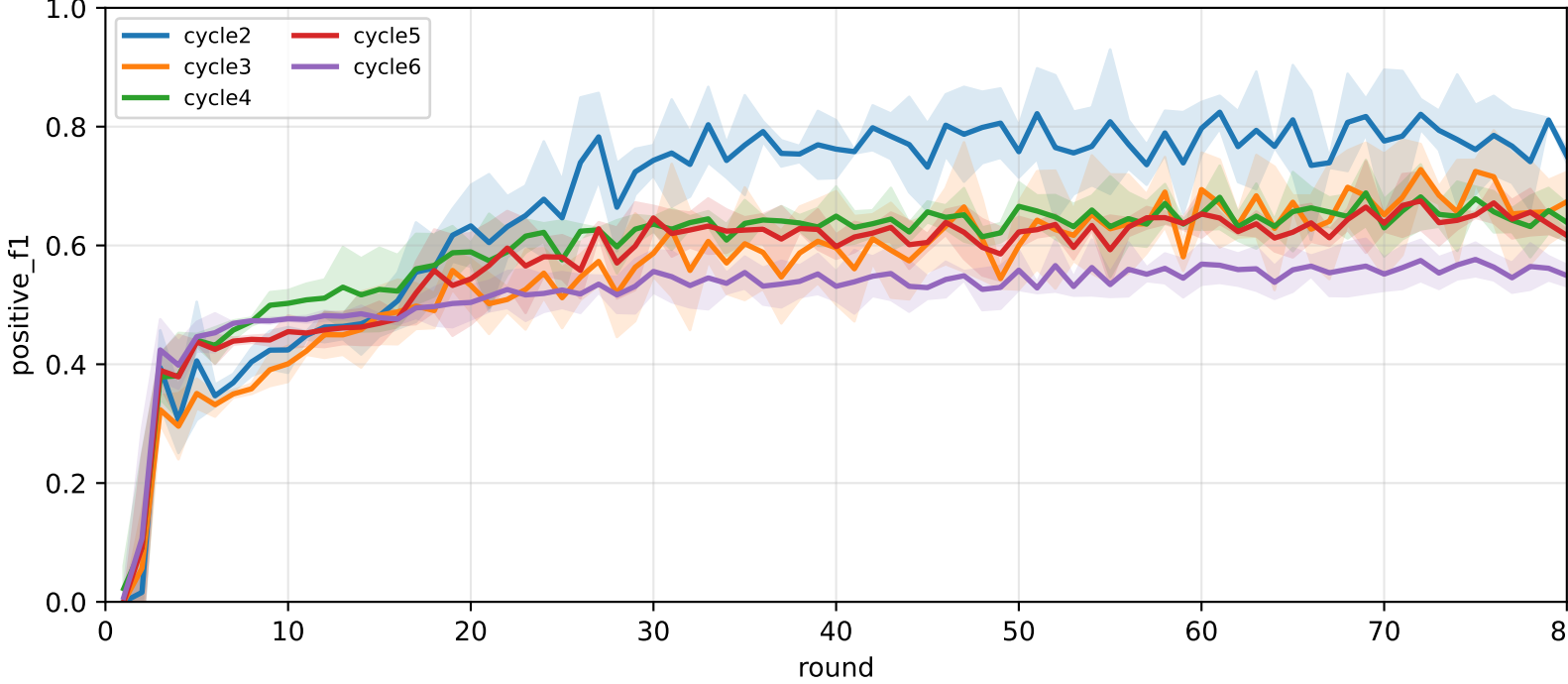
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



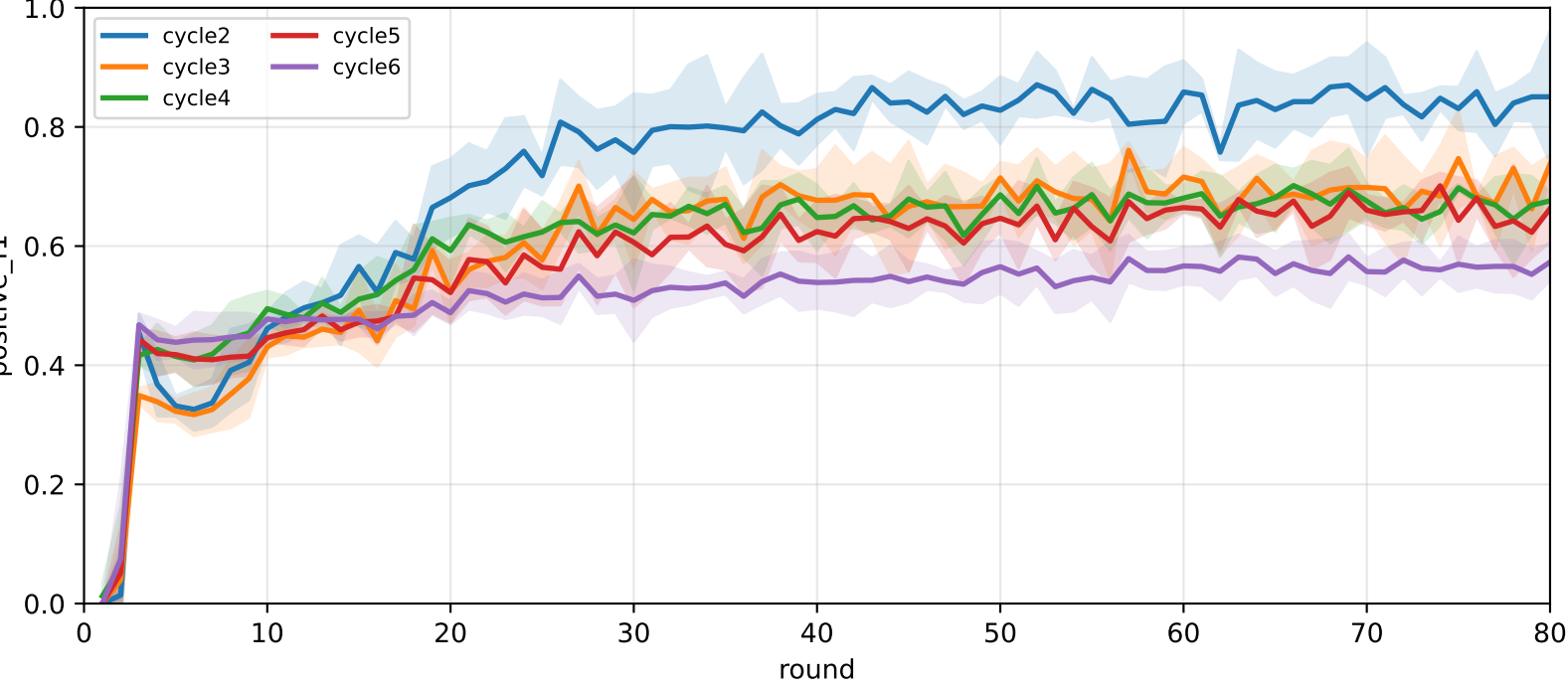
Pooled local centralized positive F1



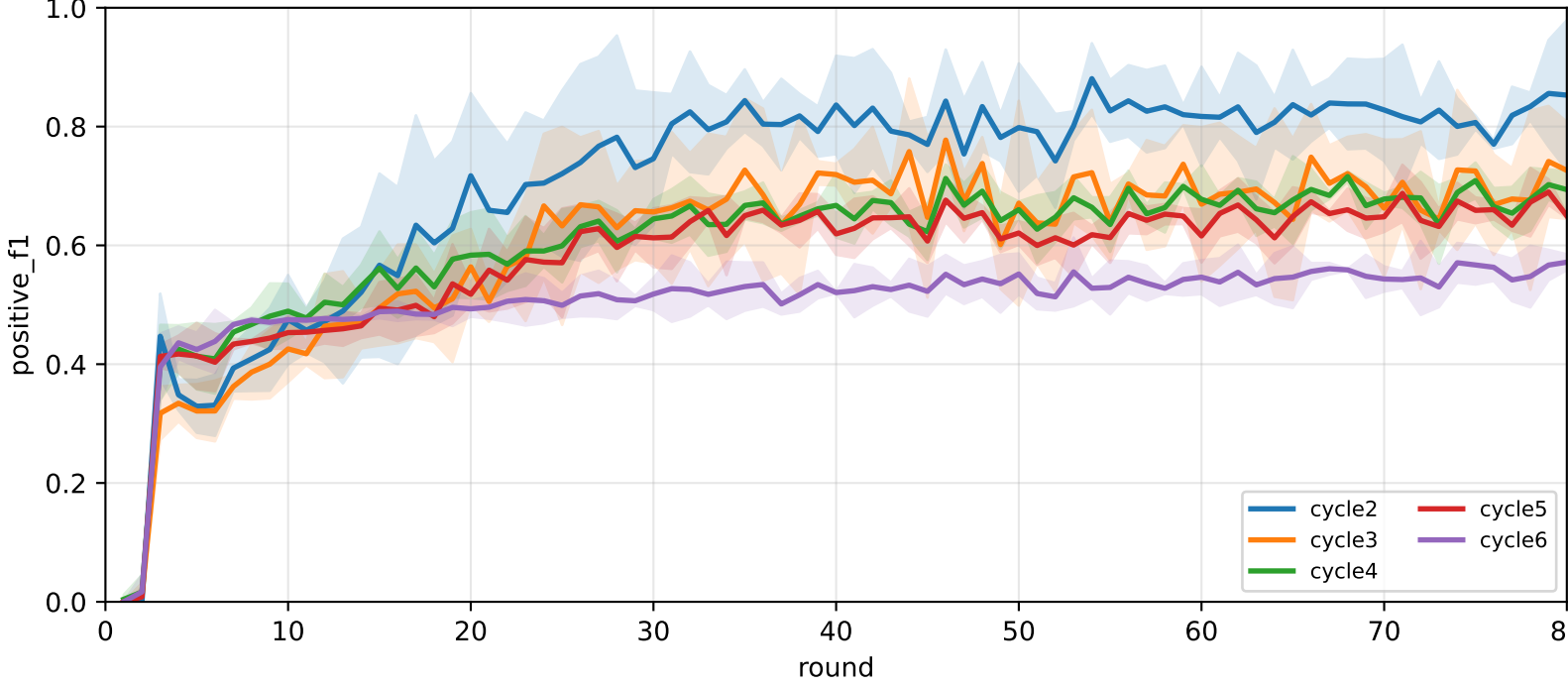
Pooled FedAvg positive F1



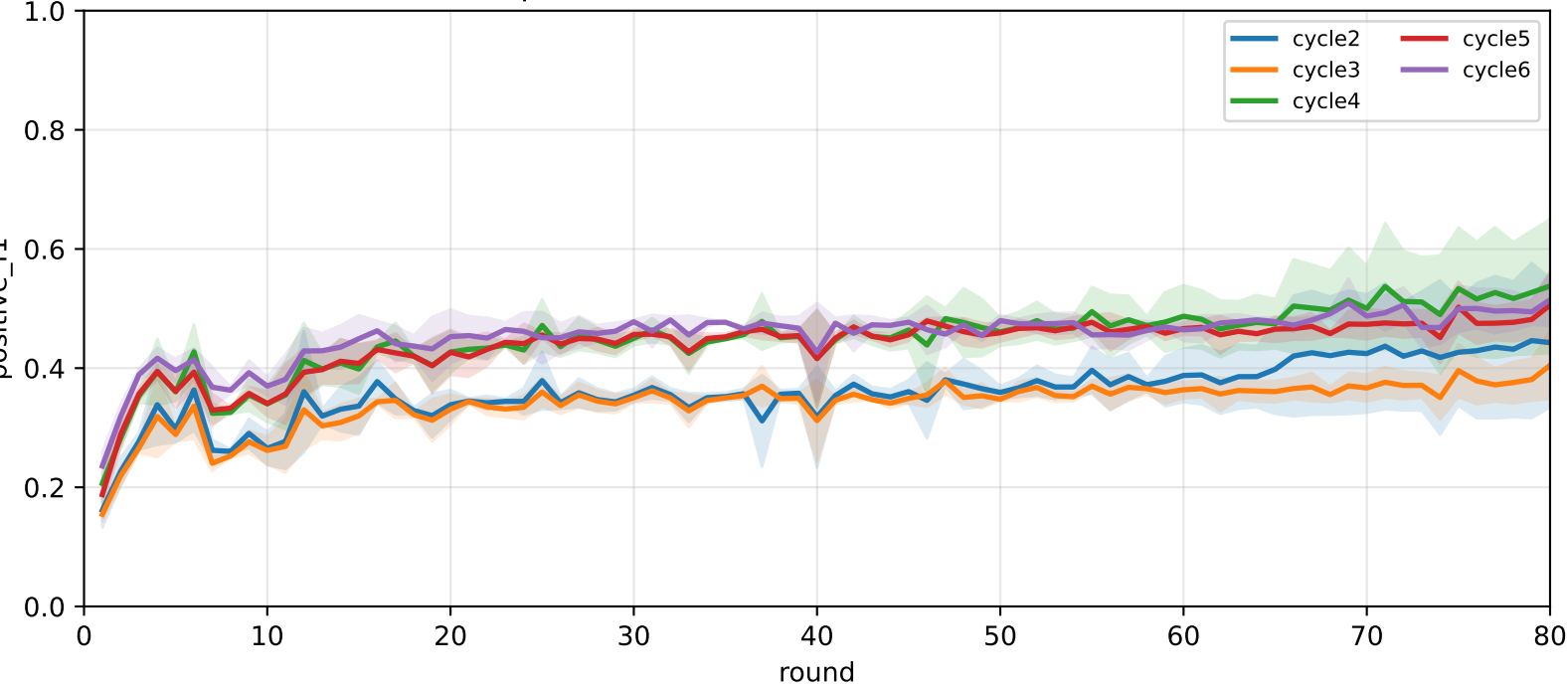
Pooled FedProx positive F1



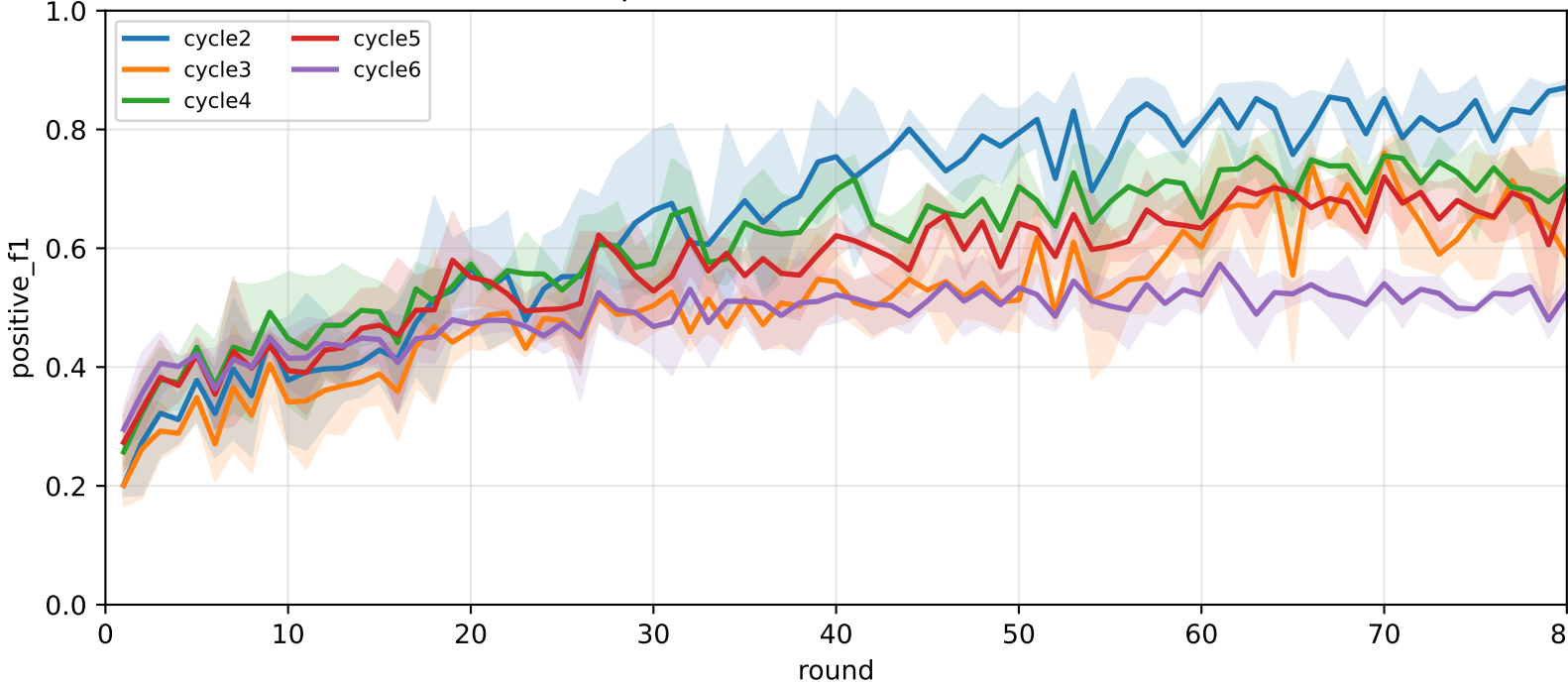
Pooled GCFL+ positive F1



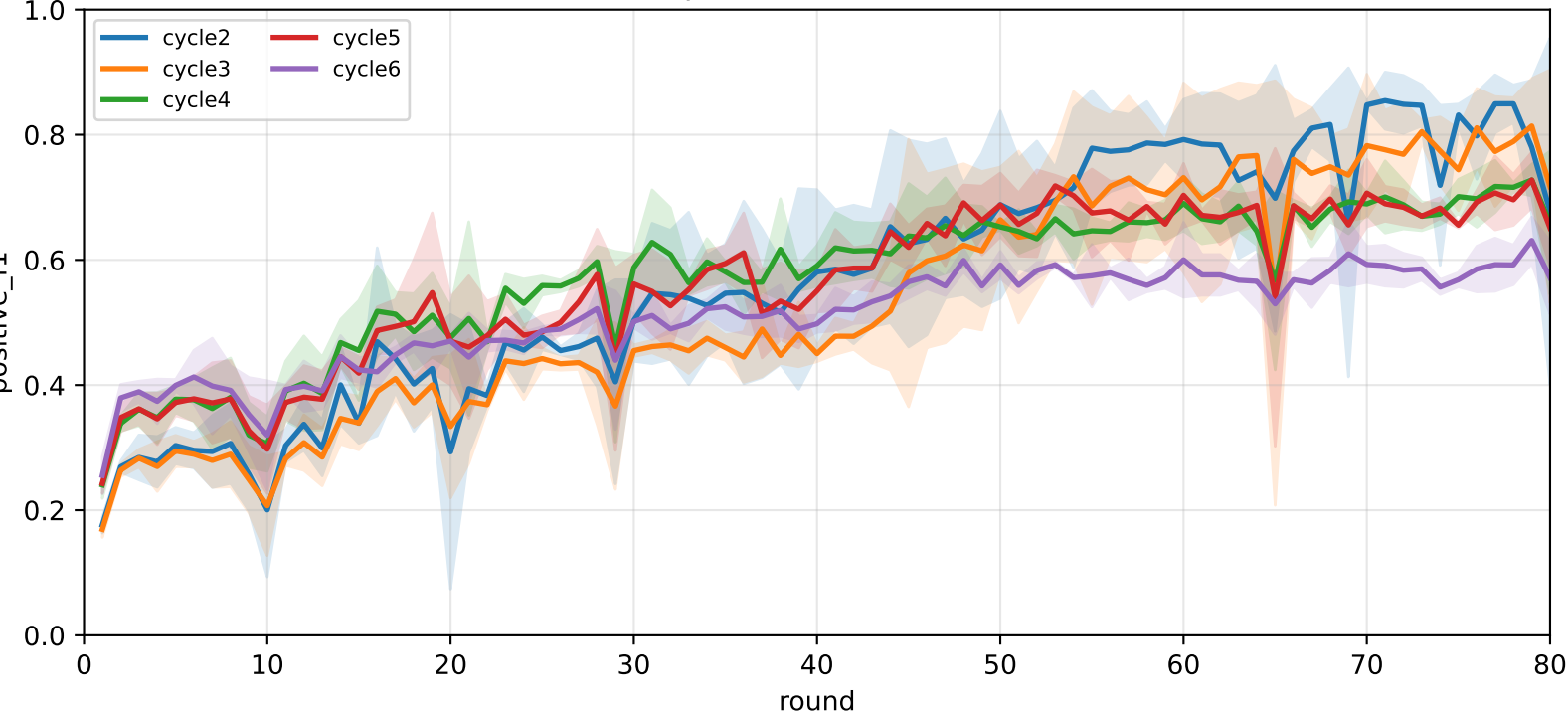
Pooled APPLE positive F1 (after-local selected model, rounds=80)



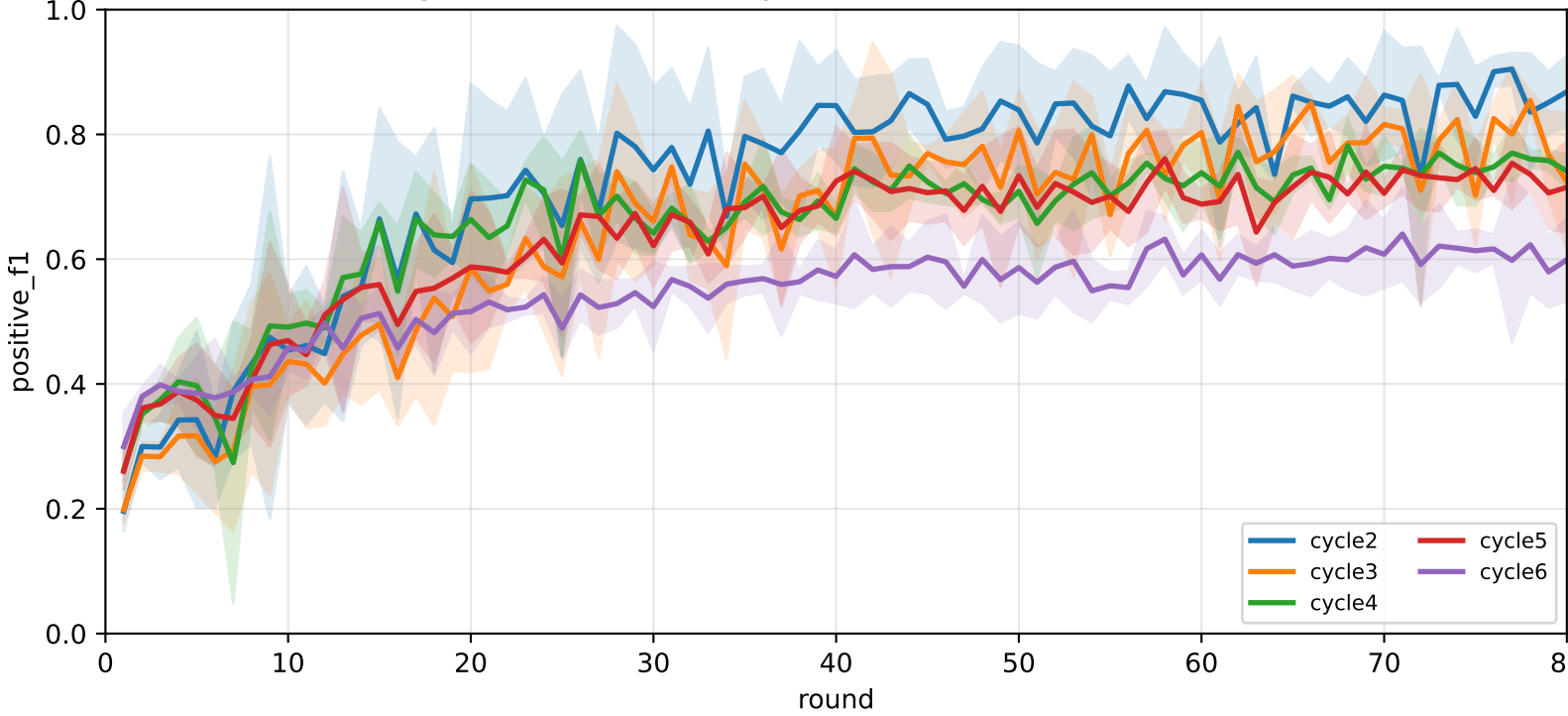
Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)



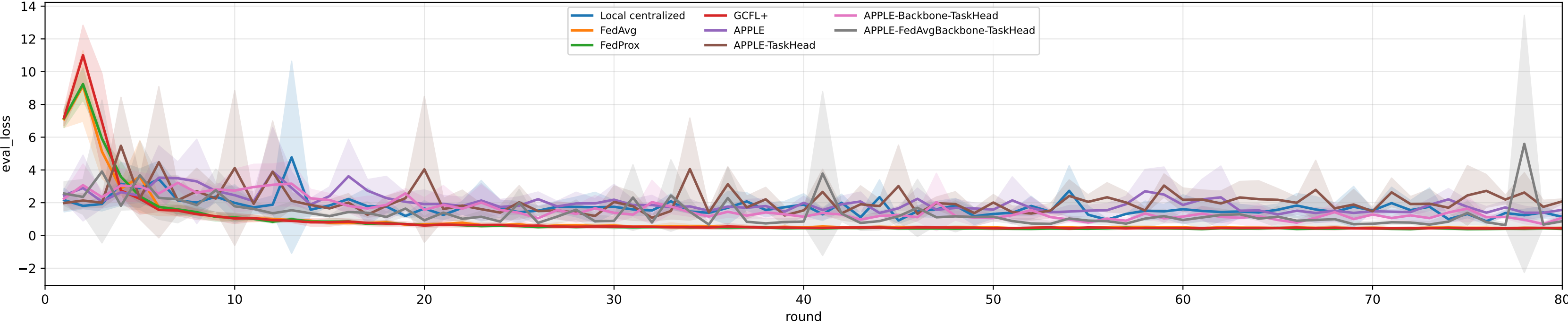
Controlled q-label heterogeneity: Mid | family aggregated / pooled performance |
family=q_task_cycle4 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% |
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

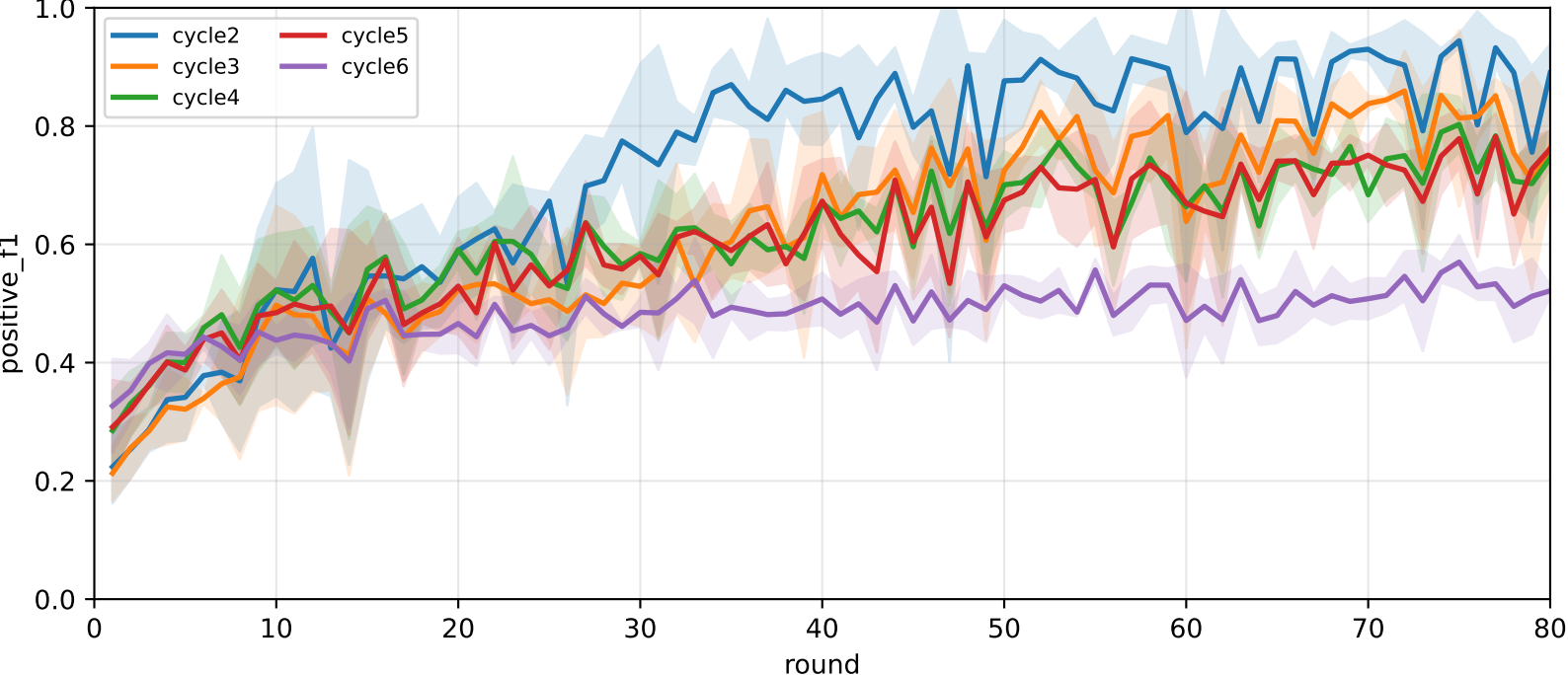
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	57.1% ± 7.9%	59.7% ± 8.6%	71.0% ± 8.9%	68.6% ± 10.4%	60.3% ± 10.1%	53.0% ± 10.3%	45.6% ± 4.2%
FedAvg	63.9% ± 1.7%	66.4% ± 1.9%	80.0% ± 5.5%	74.9% ± 4.9%	63.4% ± 1.7%	60.9% ± 1.1%	52.8% ± 2.6%
FedProx	66.2% ± 1.4%	68.6% ± 1.0%	83.7% ± 1.0%	74.3% ± 4.2%	66.1% ± 0.9%	64.1% ± 0.9%	54.9% ± 2.6%
GCFL+	65.9% ± 3.9%	68.6% ± 3.3%	83.2% ± 4.5%	76.1% ± 7.1%	67.6% ± 3.2%	62.1% ± 4.6%	54.0% ± 4.3%
APPLE	45.9% ± 3.1%	45.8% ± 3.1%	41.4% ± 5.6%	46.2% ± 2.7%	46.9% ± 3.2%	46.0% ± 2.9%	48.7% ± 2.7%
APPLE-TaskHead	54.9% ± 2.6%	55.8% ± 3.0%	59.1% ± 7.7%	60.5% ± 2.7%	52.3% ± 2.6%	56.5% ± 6.2%	50.4% ± 0.1%
APPLE-Backbone-TaskHead	65.7% ± 6.3%	67.7% ± 6.5%	79.6% ± 5.4%	72.8% ± 12.7%	64.5% ± 8.0%	67.0% ± 4.0%	54.4% ± 3.4%
APPLE-FedAvgBackbone-TaskHead	68.1% ± 3.0%	71.1% ± 2.8%	84.8% ± 3.3%	80.6% ± 7.1%	65.8% ± 1.0%	69.6% ± 4.7%	54.8% ± 3.9%

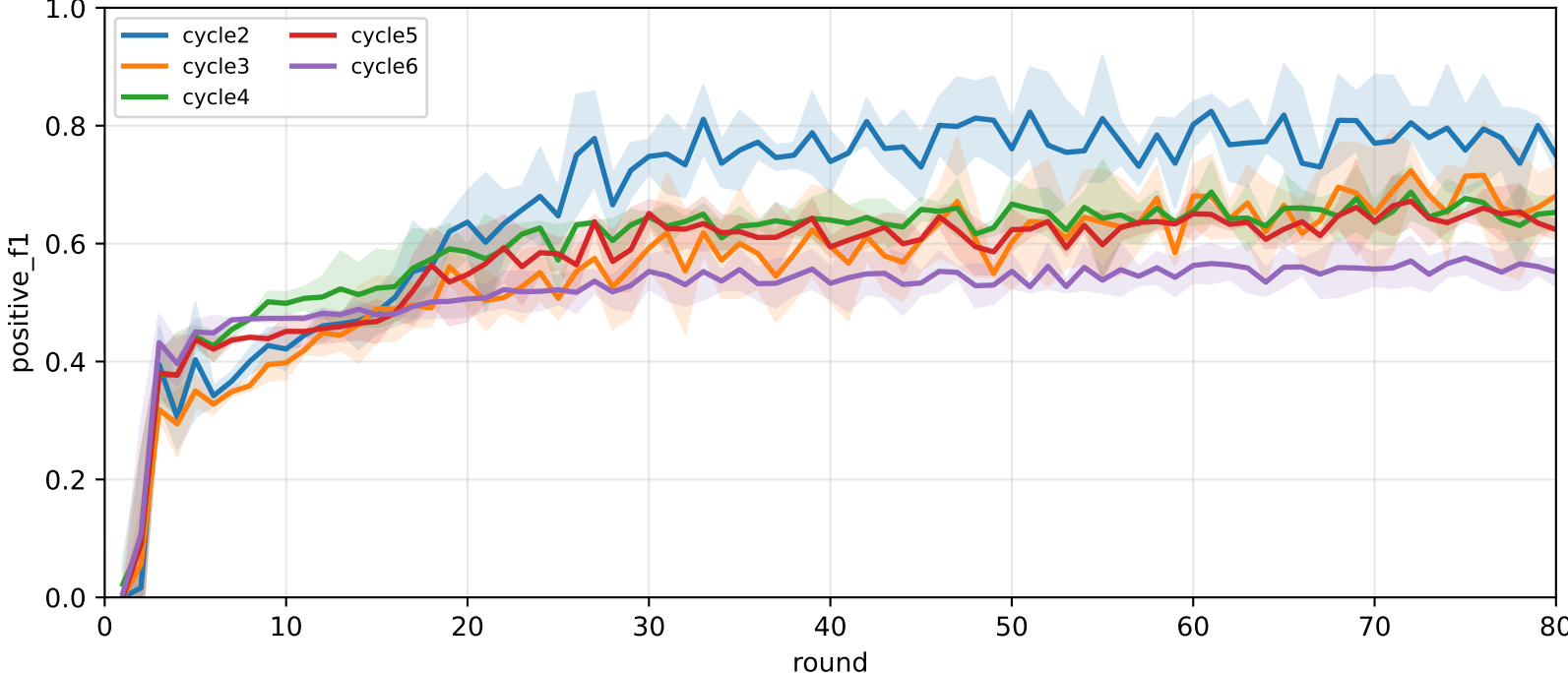
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



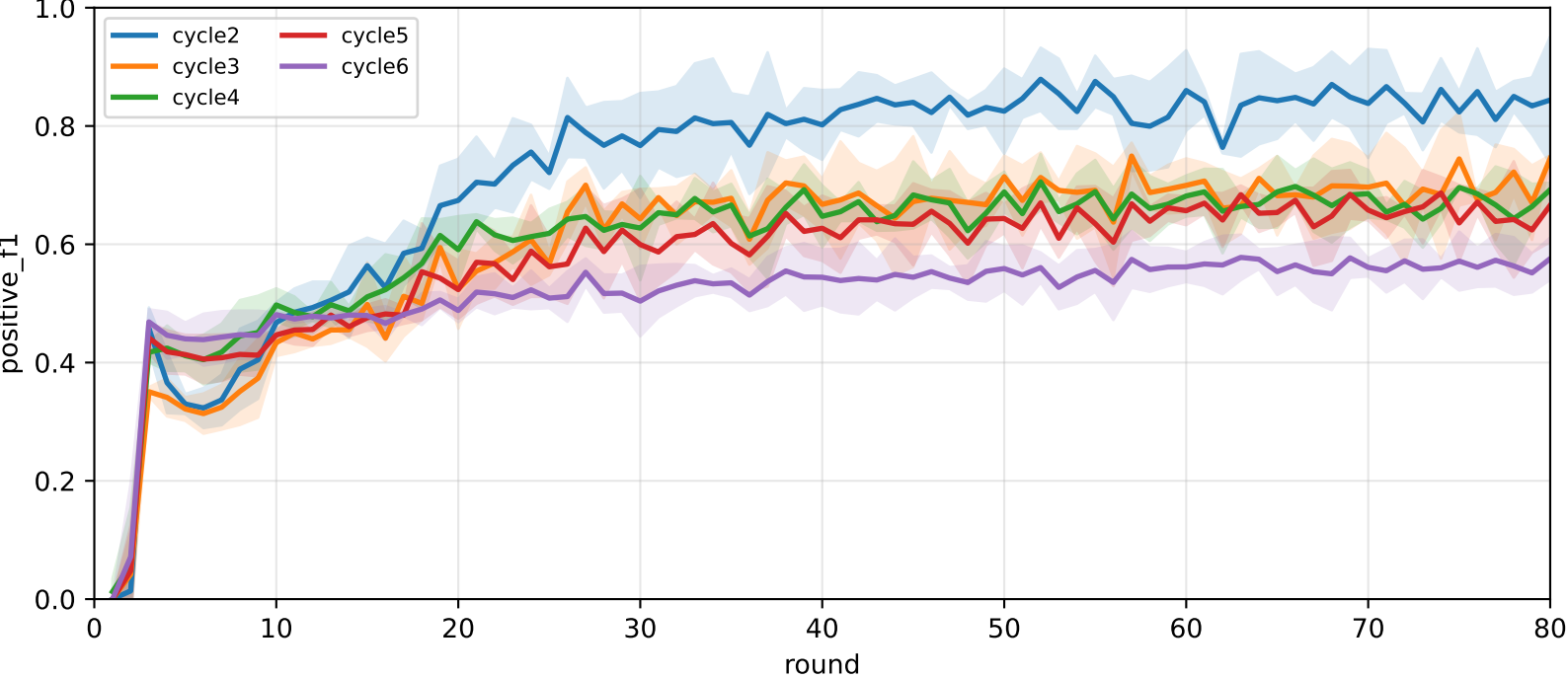
Pooled local centralized positive F1



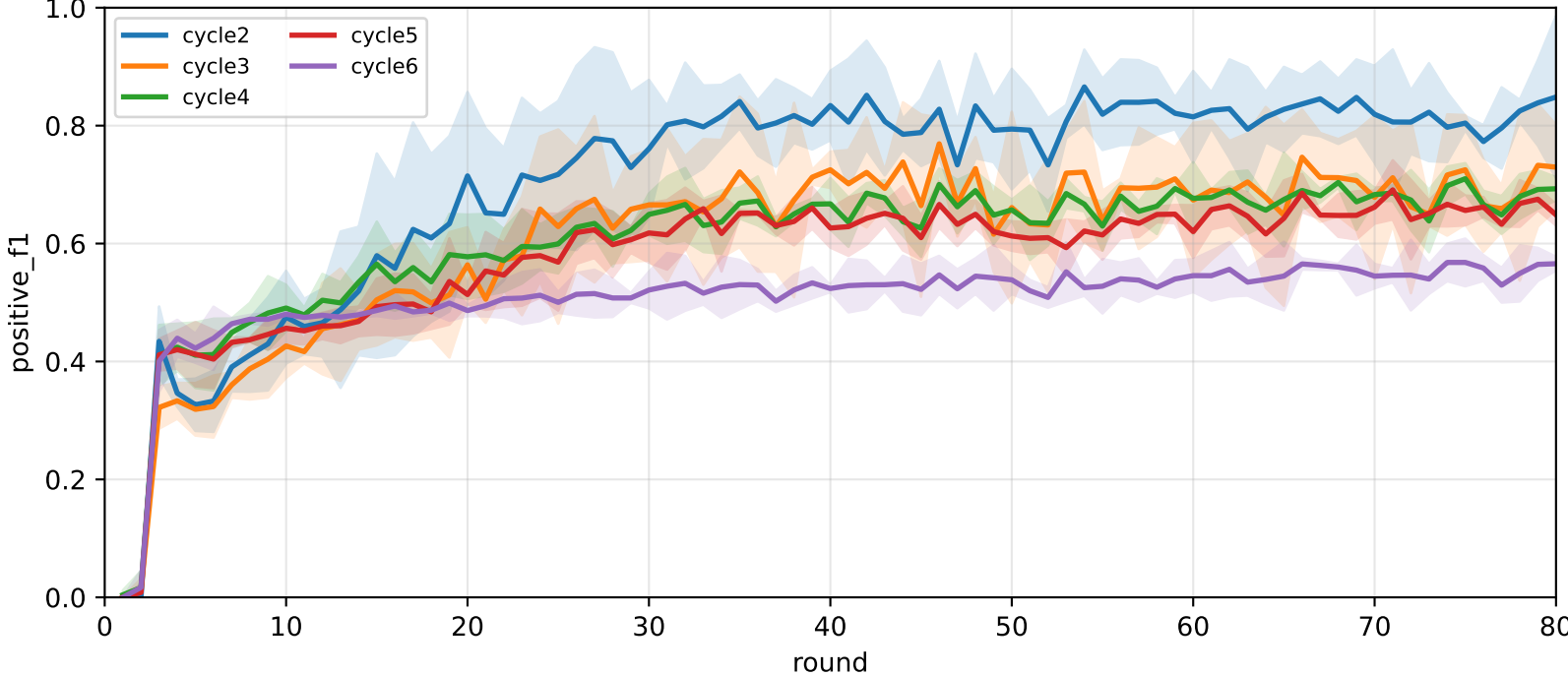
Pooled FedAvg positive F1



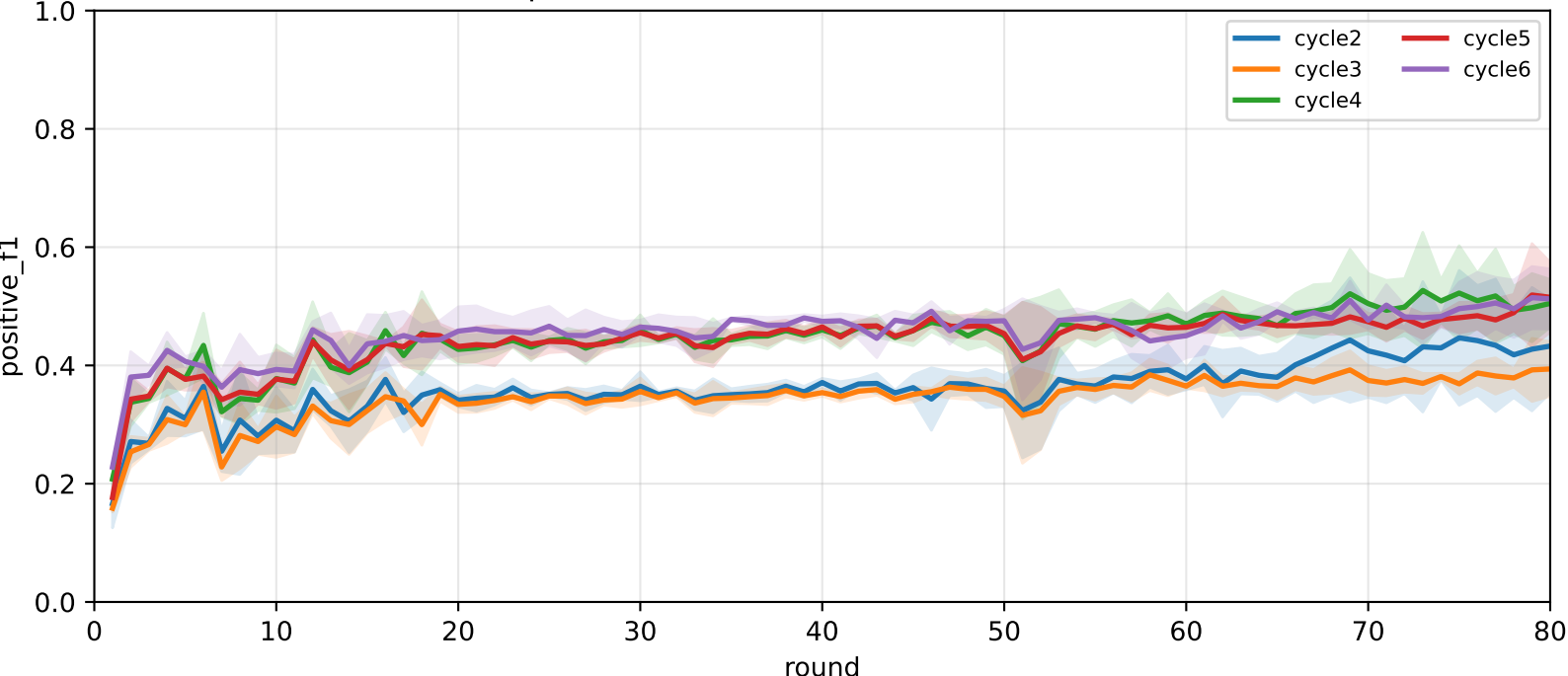
Pooled FedProx positive F1



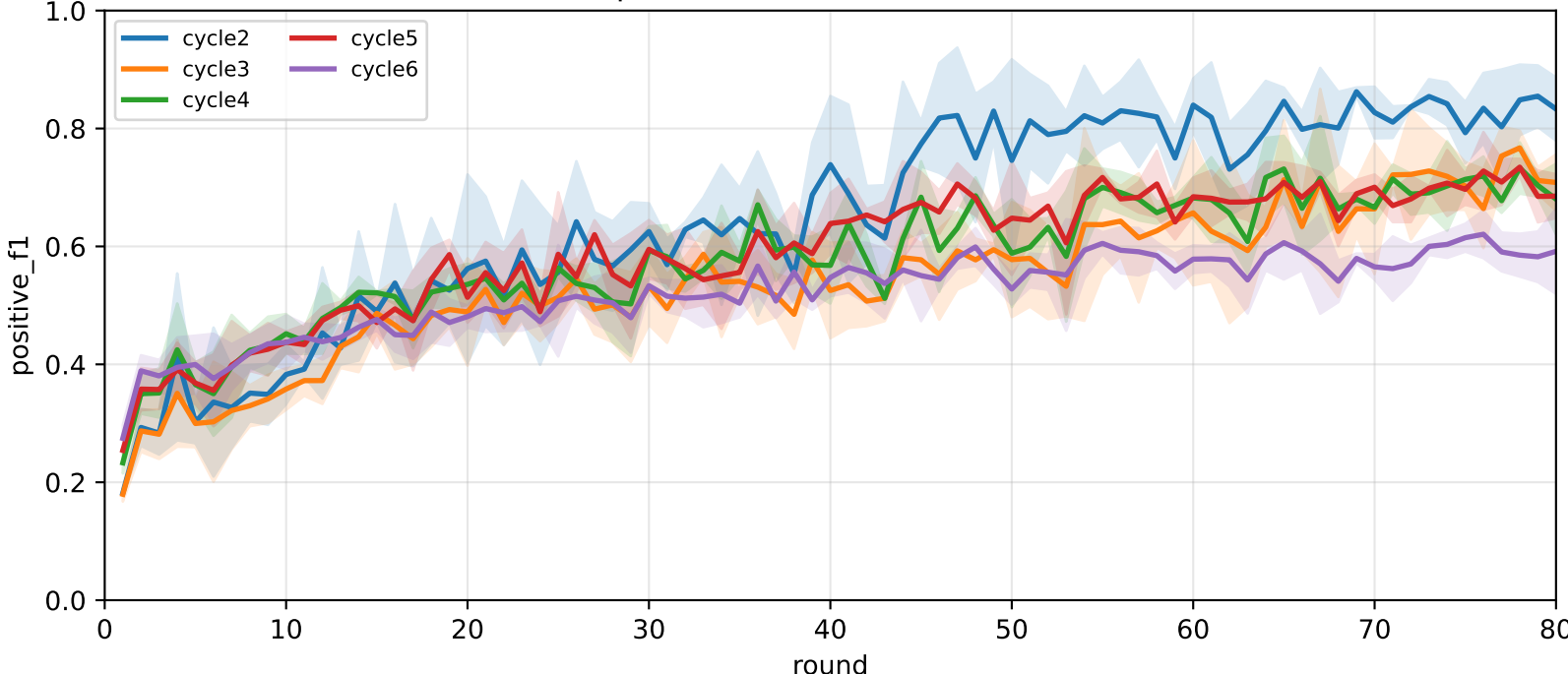
Pooled GCFL+ positive F1



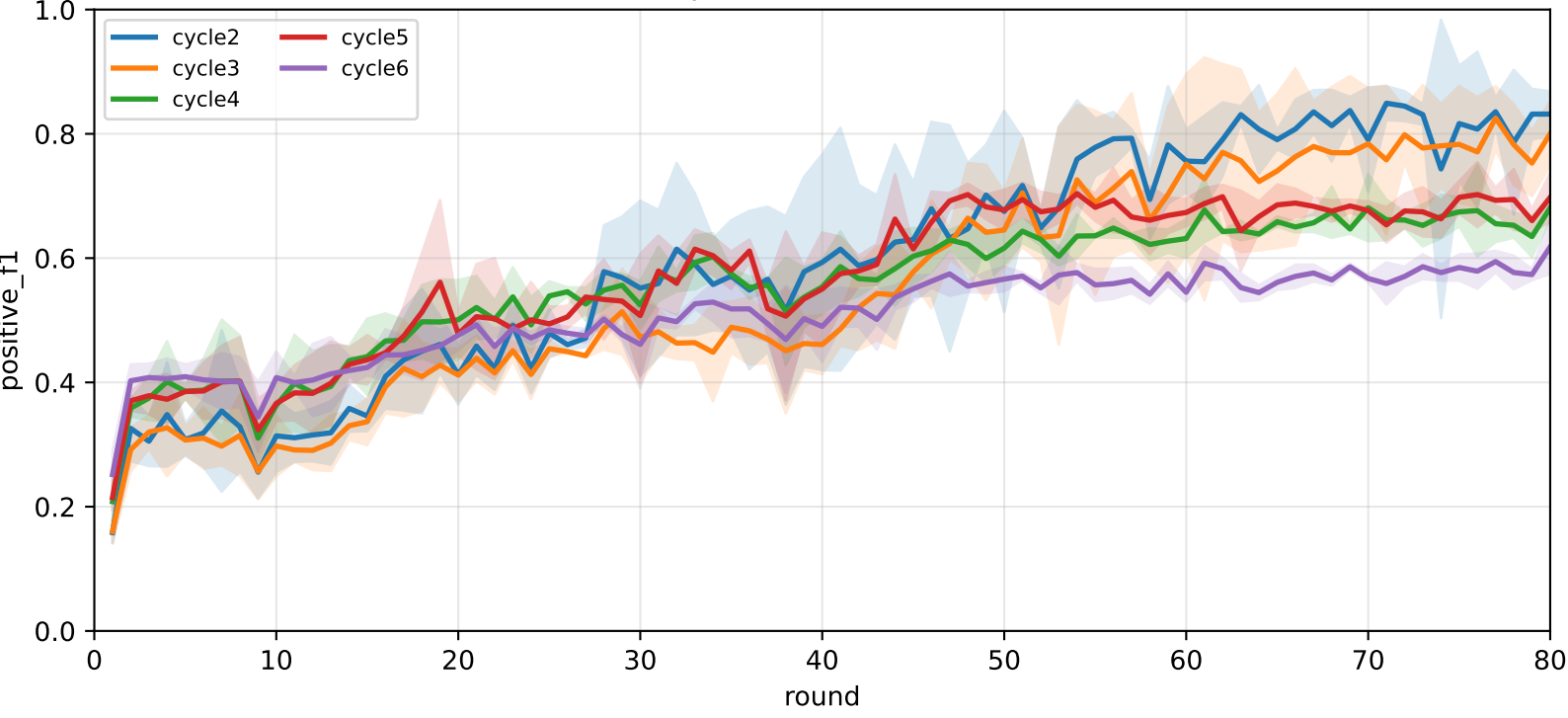
Pooled APPLE positive F1 (after-local selected model, rounds=80)



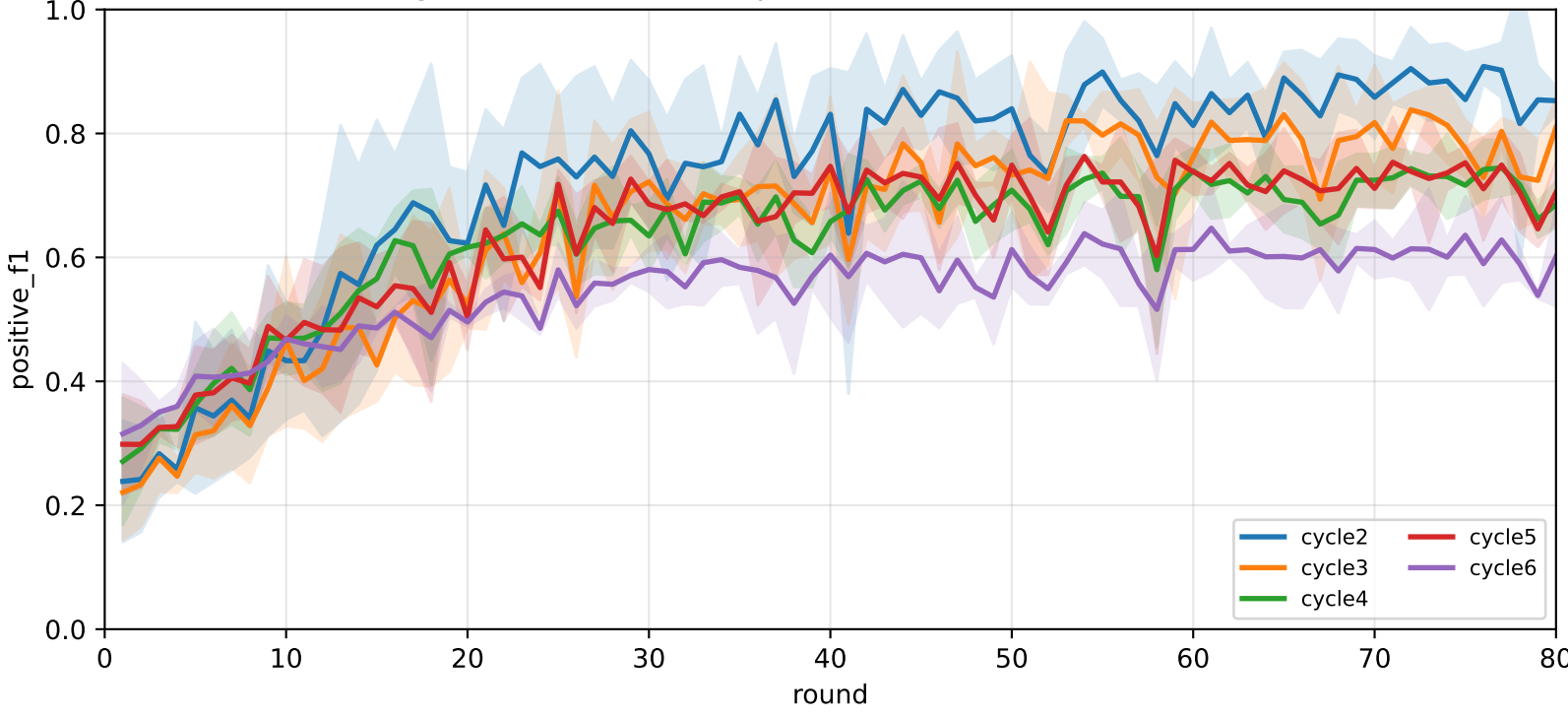
Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)



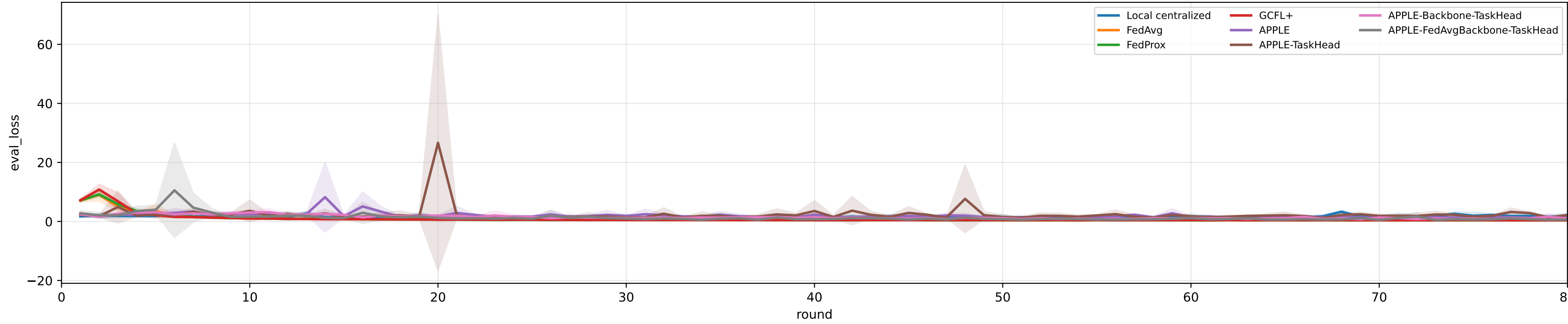
Controlled q-label heterogeneity: Mid | family aggregated / pooled performance |
family=q_task_cycle5 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |
cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

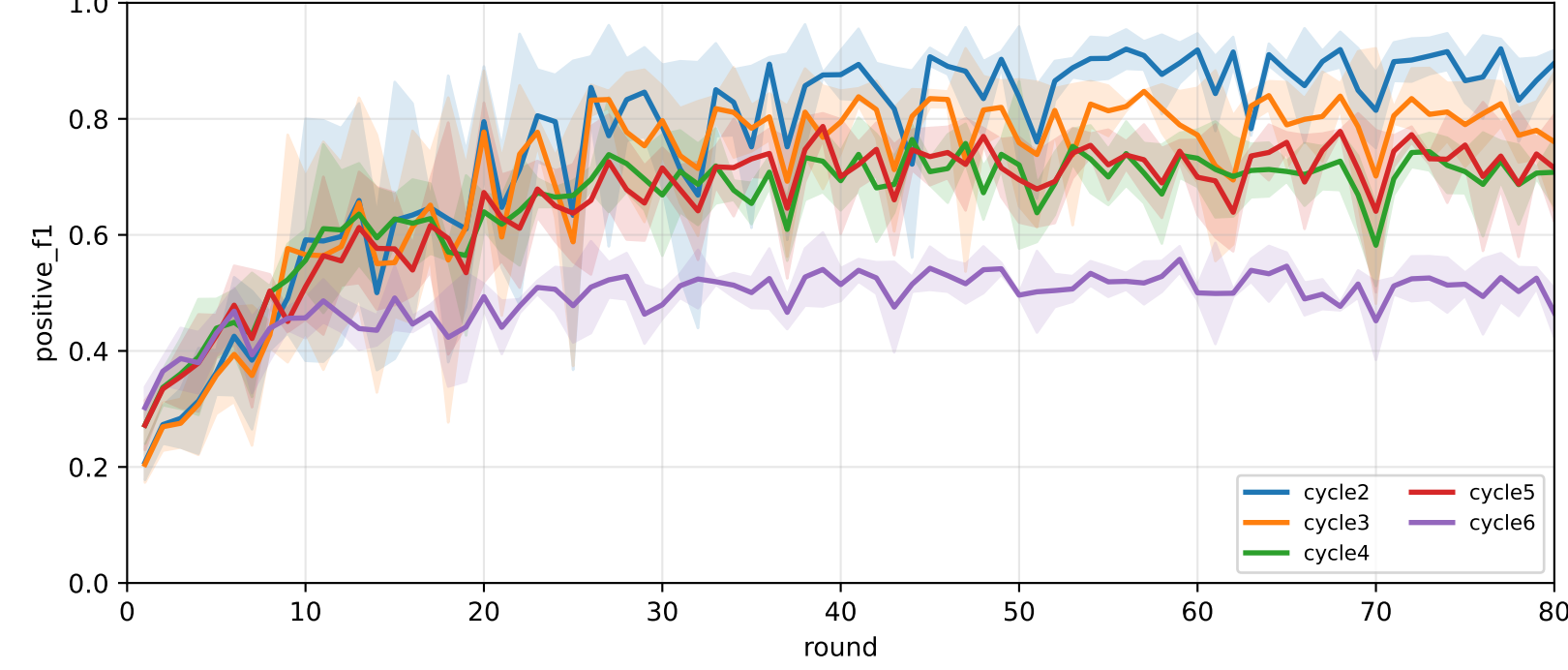
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	62.7% ± 2.4%	65.2% ± 2.6%	76.5% ± 5.8%	75.7% ± 4.0%	59.8% ± 2.9%	64.6% ± 3.0%	49.4% ± 2.2%
FedAvg	63.4% ± 1.5%	65.9% ± 1.5%	79.5% ± 4.2%	74.0% ± 4.7%	63.0% ± 1.6%	60.0% ± 0.4%	52.9% ± 2.5%
FedProx	65.9% ± 1.1%	68.2% ± 0.8%	82.6% ± 2.5%	73.4% ± 4.0%	66.3% ± 0.9%	63.6% ± 0.3%	55.2% ± 2.4%
GCFL+	65.9% ± 4.3%	68.5% ± 3.8%	82.1% ± 3.0%	75.7% ± 7.9%	67.7% ± 4.4%	63.2% ± 4.9%	53.9% ± 4.4%
APPLE	47.5% ± 6.3%	47.5% ± 6.5%	42.9% ± 9.1%	48.6% ± 6.5%	50.8% ± 9.3%	46.6% ± 5.5%	48.5% ± 2.3%
APPLE-TaskHead	44.6% ± 5.1%	46.0% ± 6.4%	55.7% ± 18.6%	46.5% ± 6.7%	42.1% ± 3.5%	41.5% ± 2.5%	44.0% ± 2.9%
APPLE-Backbone-TaskHead	65.6% ± 1.8%	67.1% ± 2.1%	72.5% ± 3.0%	79.4% ± 5.6%	63.4% ± 0.7%	63.8% ± 2.6%	56.6% ± 2.0%
APPLE-FedAvgBackbone-TaskHead	72.1% ± 1.2%	74.6% ± 1.3%	87.4% ± 4.8%	85.5% ± 2.1%	72.8% ± 3.6%	66.6% ± 3.7%	60.5% ± 2.0%

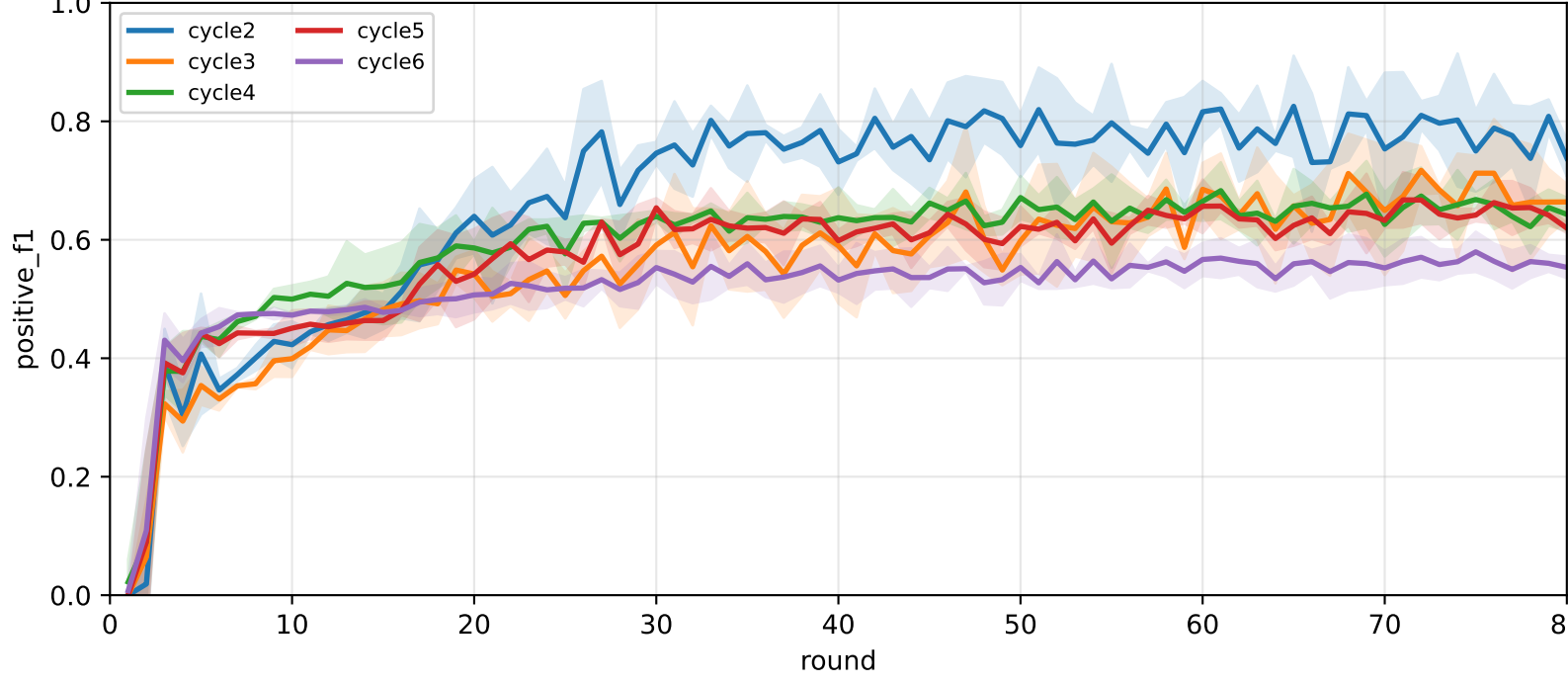
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



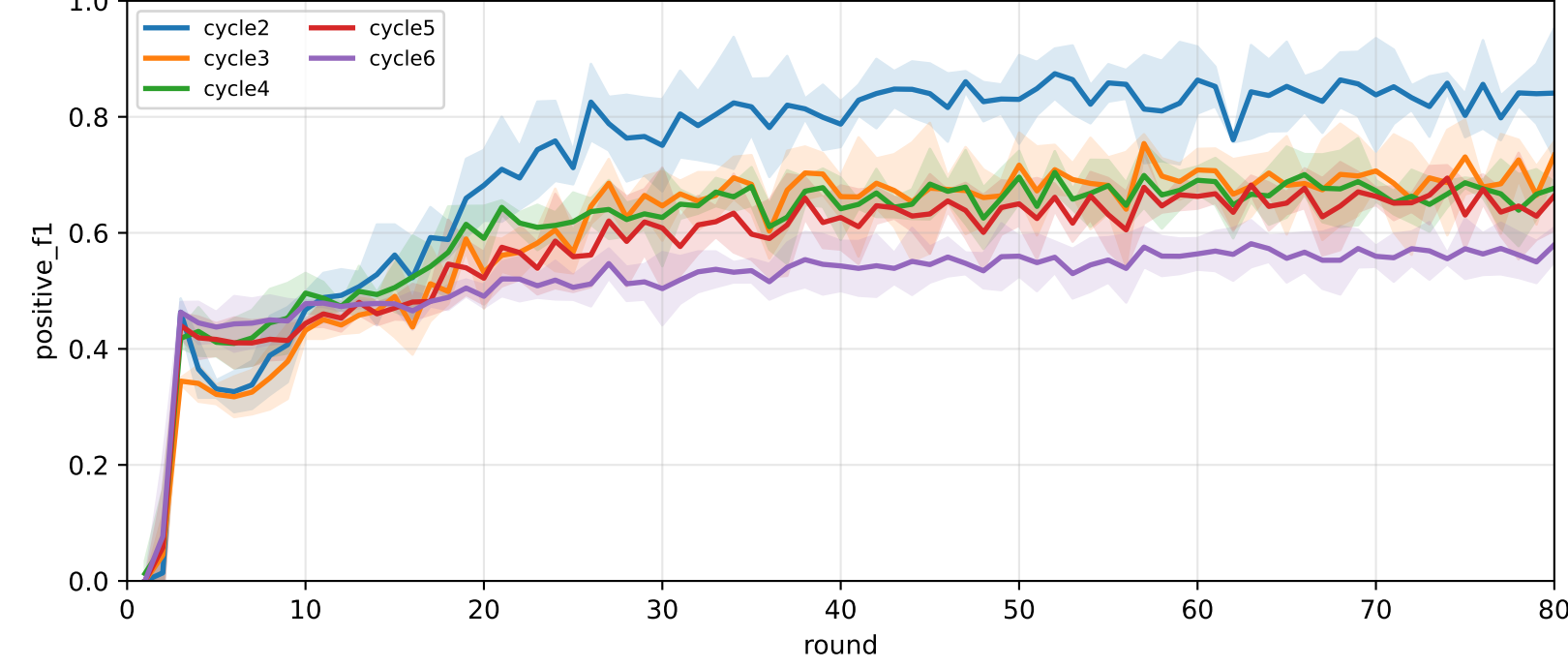
Pooled local centralized positive F1



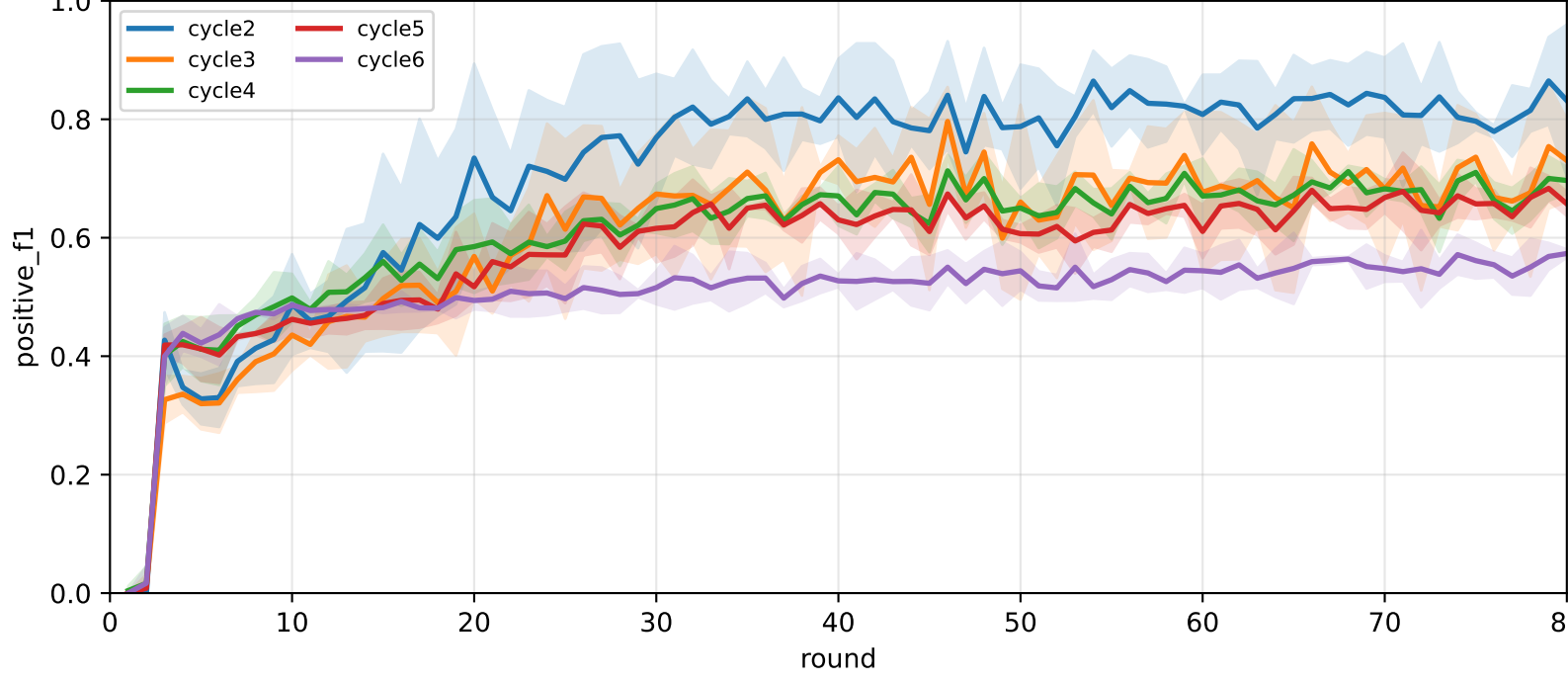
Pooled FedAvg positive F1



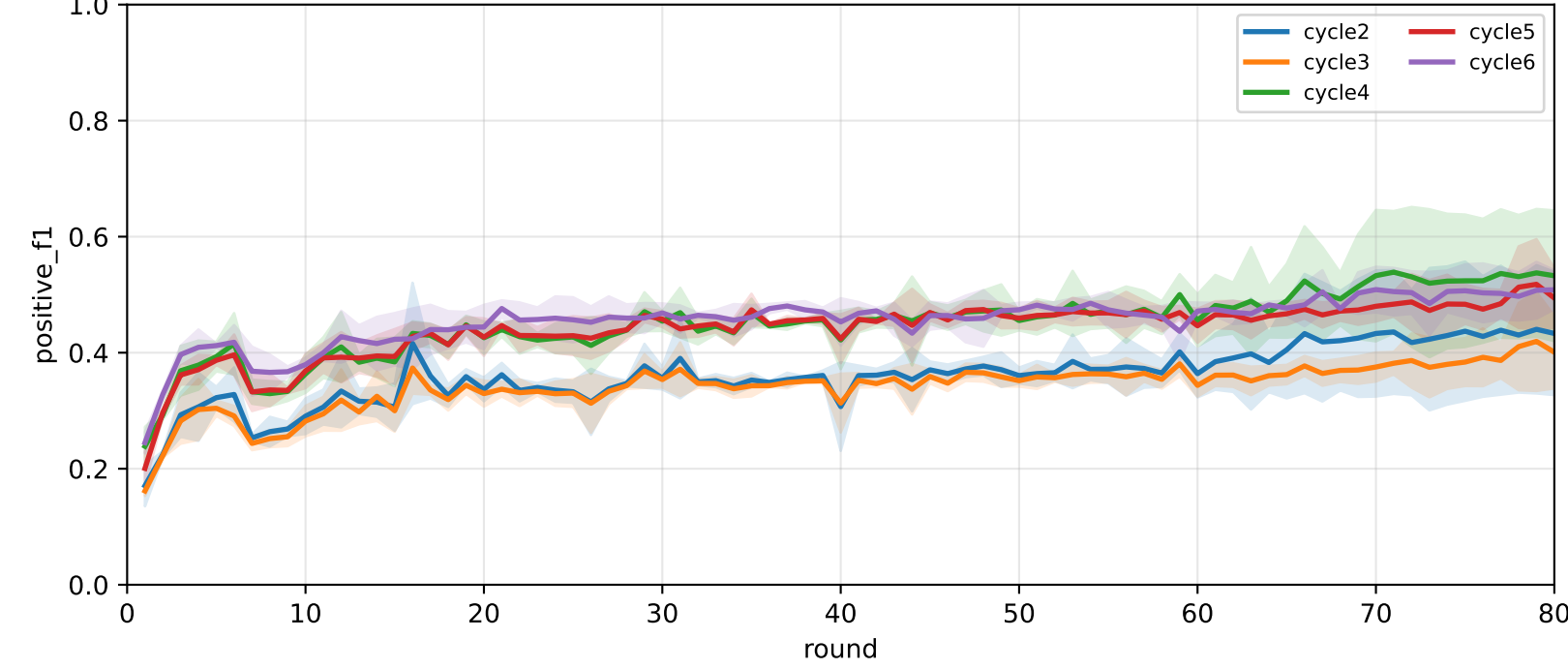
Pooled FedProx positive F1



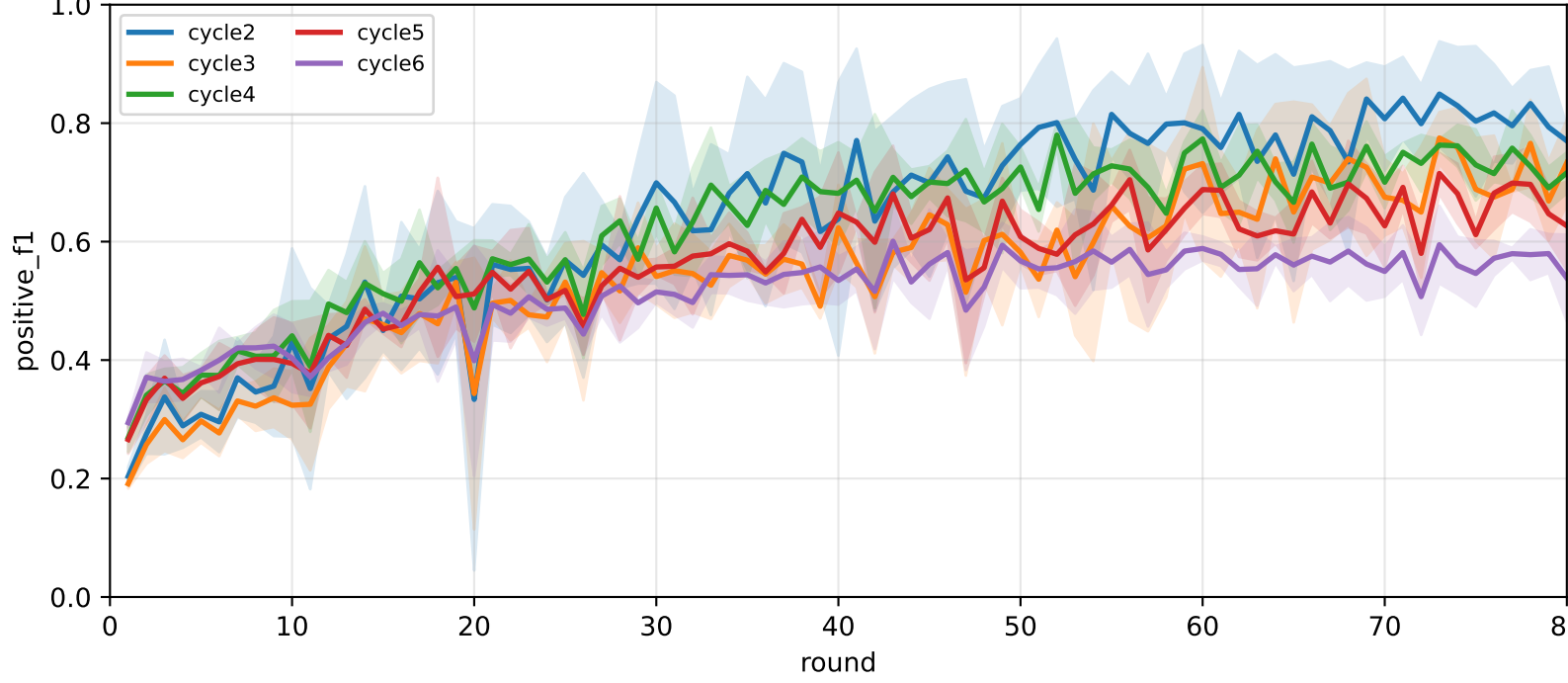
Pooled GCFL+ positive F1



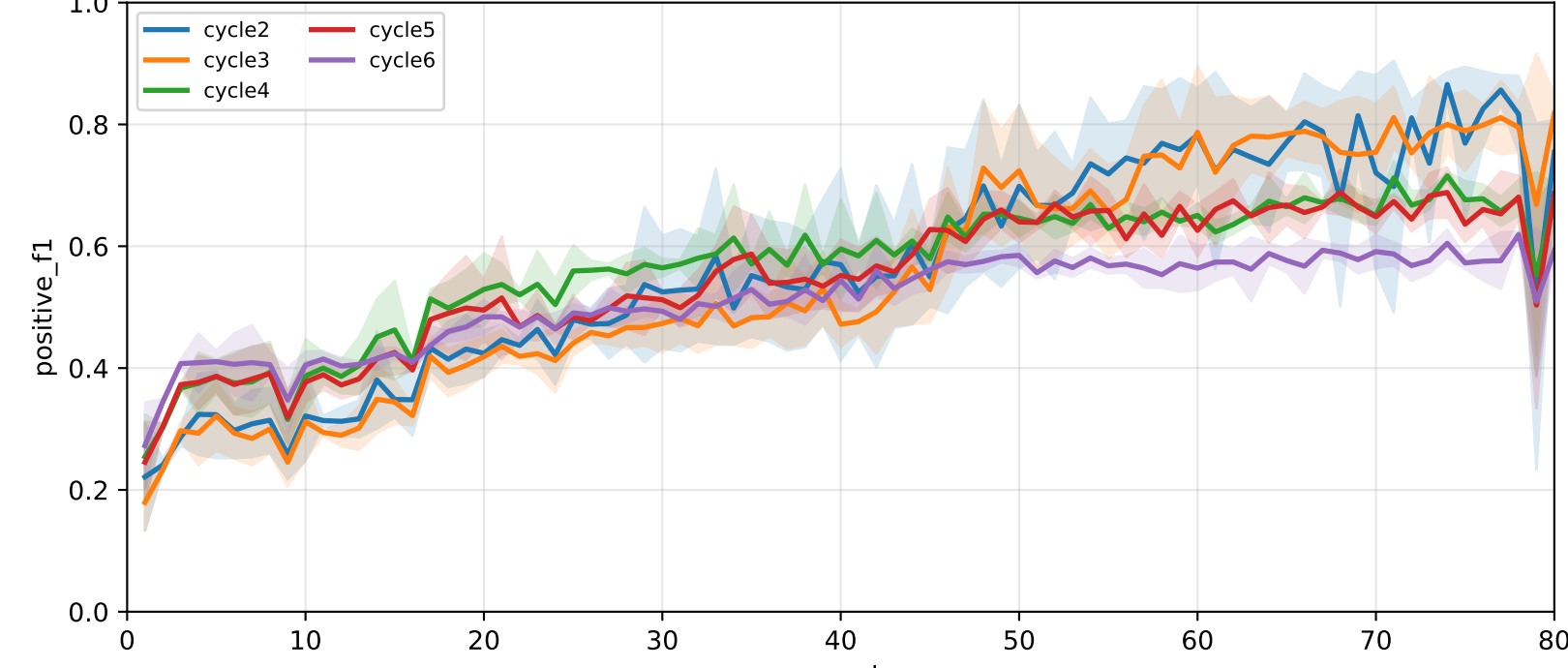
Pooled APPLE positive F1 (after-local selected model, rounds=80)



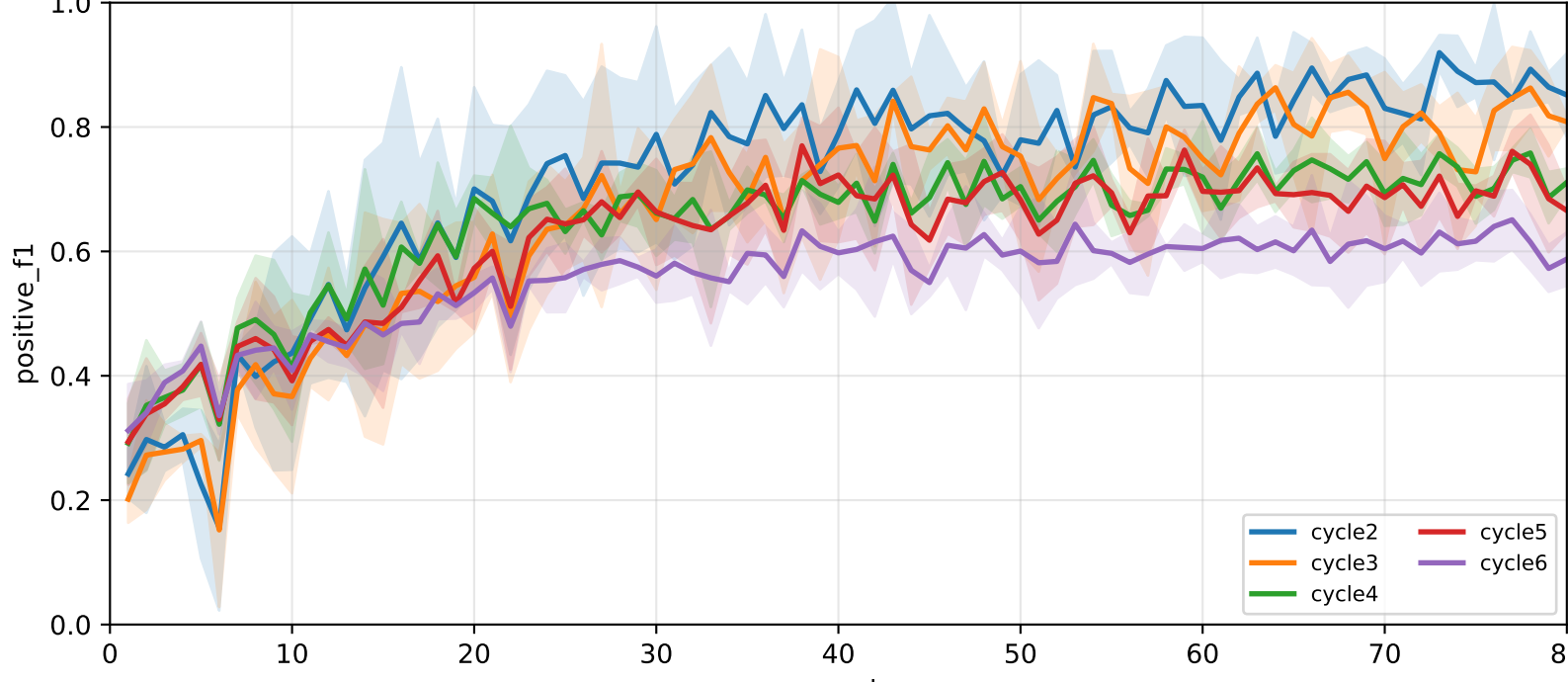
Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)



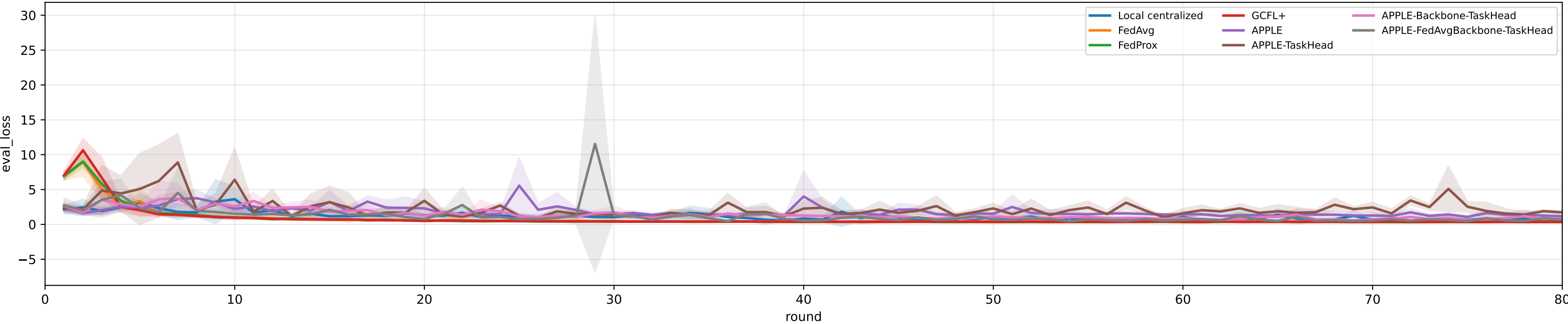
Controlled q-label heterogeneity: Mid | family aggregated / pooled performance |
family=q_task_cycle6 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

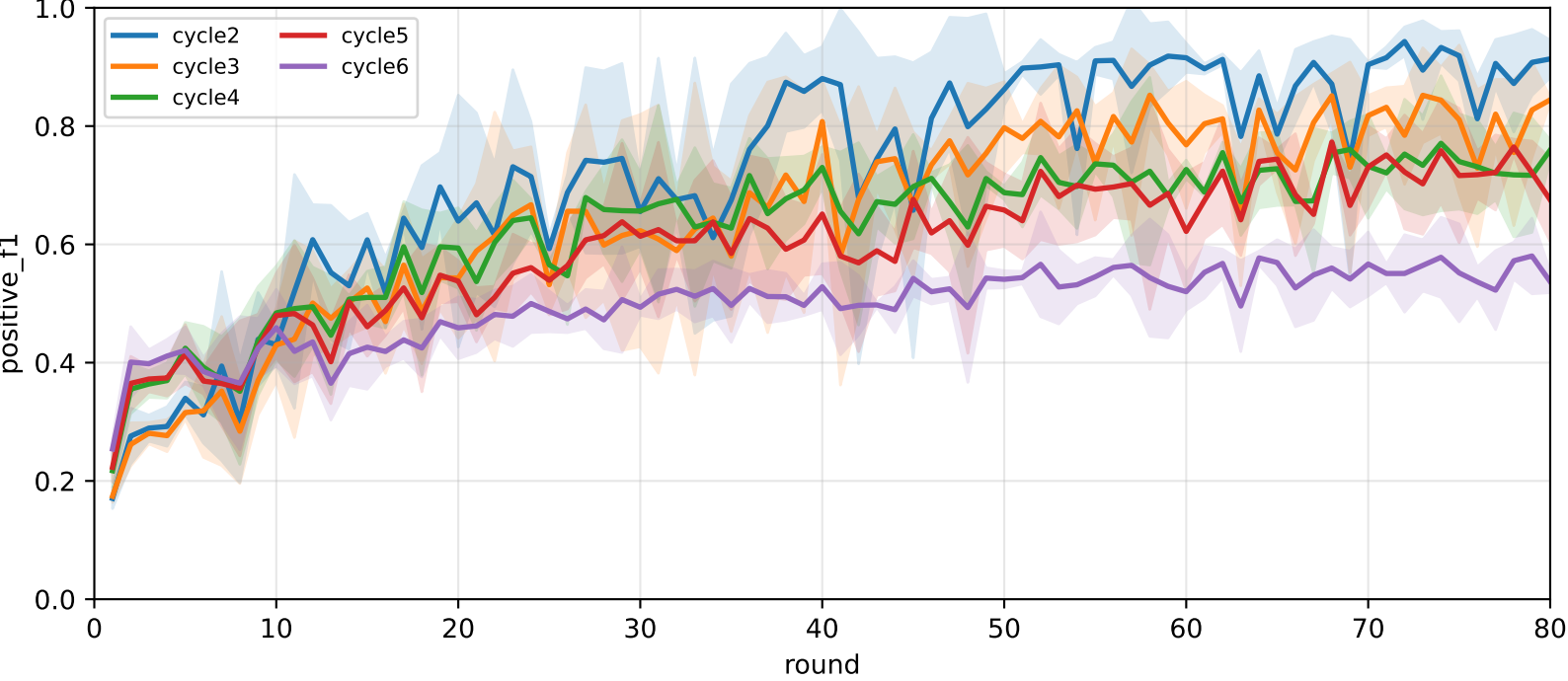
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	63.1% ± 3.5%	67.2% ± 3.7%	86.9% ± 6.6%	77.7% ± 9.8%	63.1% ± 2.3%	58.3% ± 3.3%	49.9% ± 3.9%
FedAvg	64.1% ± 1.0%	66.7% ± 1.0%	80.9% ± 5.1%	75.0% ± 5.3%	63.3% ± 1.5%	61.3% ± 0.5%	53.0% ± 2.1%
FedProx	66.6% ± 1.0%	69.1% ± 0.7%	84.1% ± 2.1%	74.0% ± 4.2%	66.8% ± 0.9%	65.3% ± 0.8%	55.1% ± 2.0%
GCFL+	66.2% ± 4.2%	68.9% ± 3.6%	83.4% ± 3.3%	76.6% ± 7.4%	68.1% ± 3.5%	62.7% ± 5.1%	53.9% ± 4.5%
APPLE	48.4% ± 5.6%	48.5% ± 5.9%	45.9% ± 10.3%	49.1% ± 4.9%	52.1% ± 8.2%	46.6% ± 4.4%	48.8% ± 2.2%
APPLE-TaskHead	55.9% ± 6.5%	57.1% ± 7.4%	60.8% ± 14.9%	61.3% ± 2.7%	58.1% ± 8.5%	56.7% ± 8.2%	48.4% ± 3.2%
APPLE-Backbone-TaskHead	67.0% ± 2.1%	69.2% ± 2.1%	80.0% ± 2.6%	79.6% ± 5.5%	64.5% ± 2.4%	65.9% ± 3.5%	55.7% ± 1.0%
APPLE-FedAvgBackbone-TaskHead	65.9% ± 1.8%	68.8% ± 0.7%	82.5% ± 5.9%	77.5% ± 3.0%	62.7% ± 3.9%	67.5% ± 3.4%	53.5% ± 3.8%

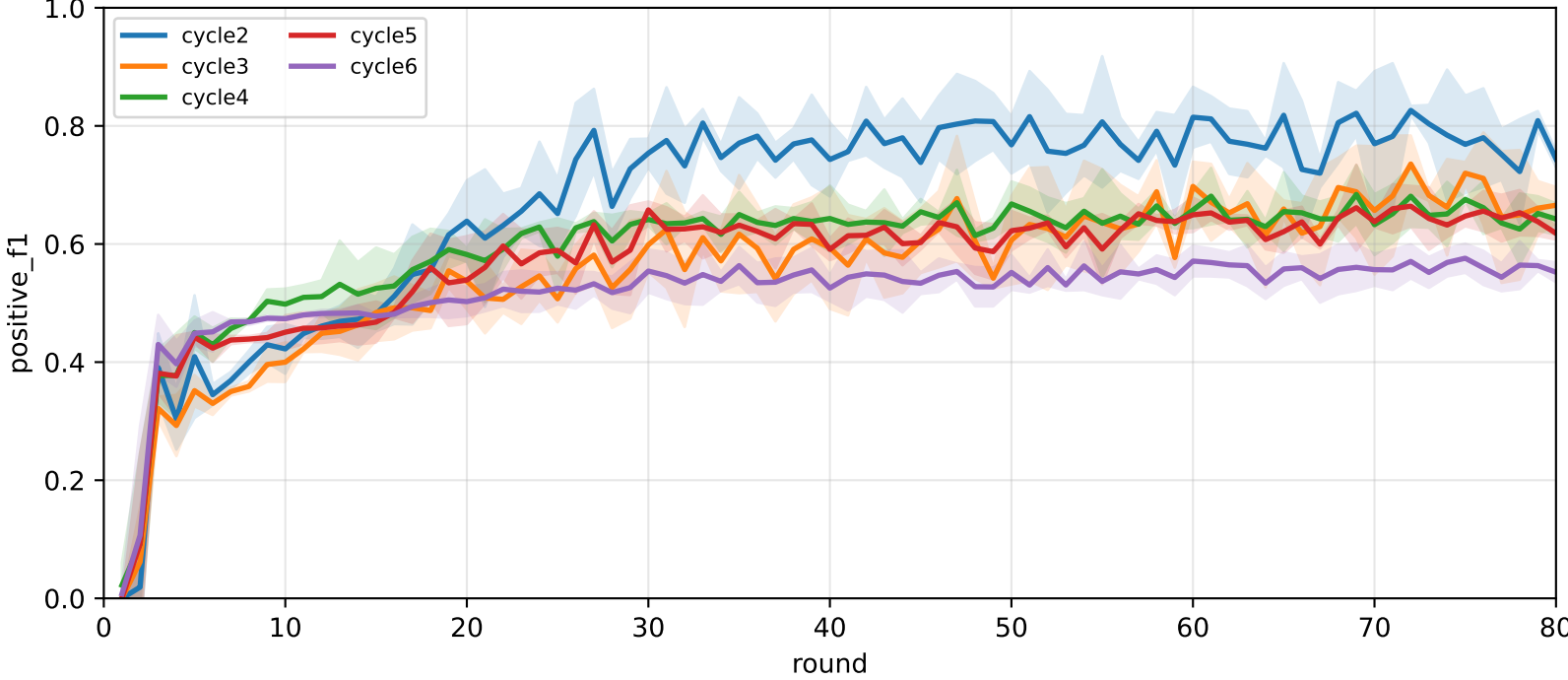
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



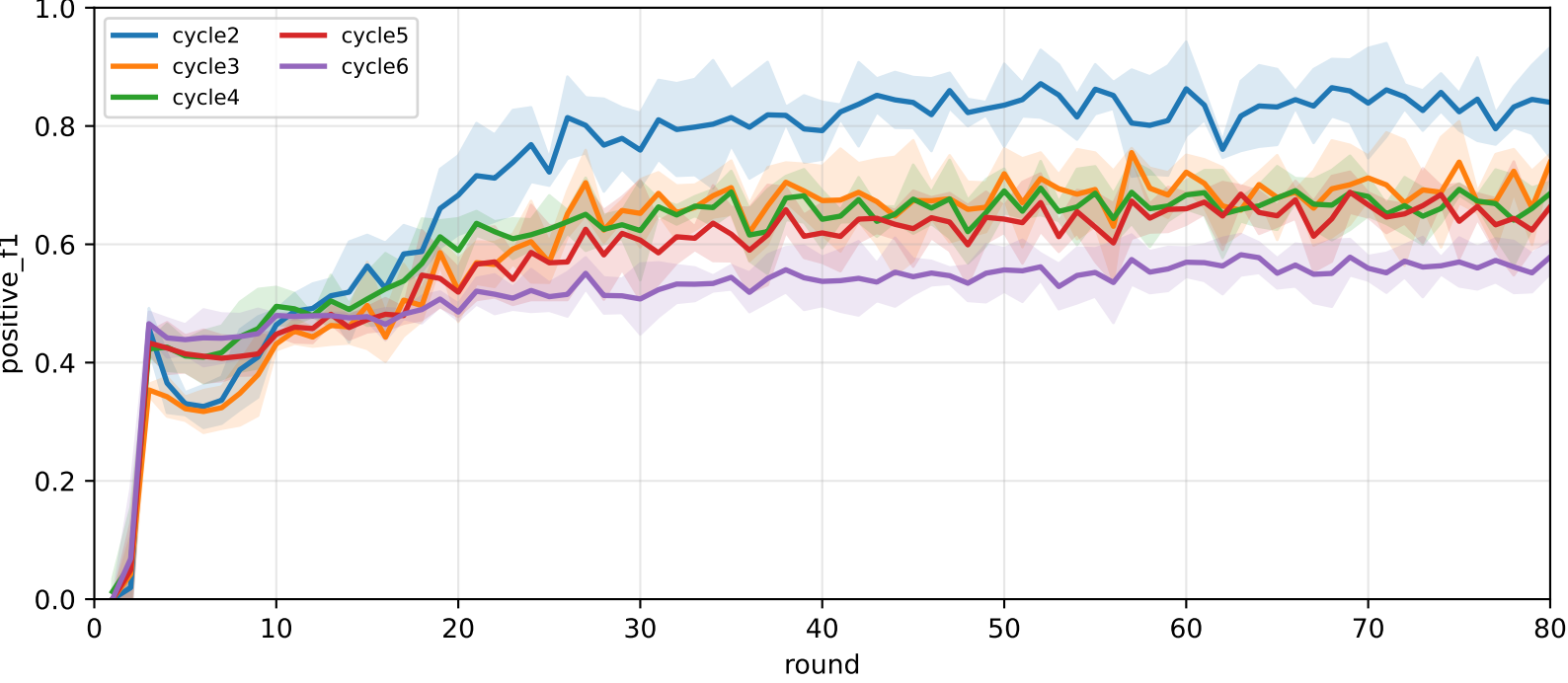
Pooled local centralized positive F1



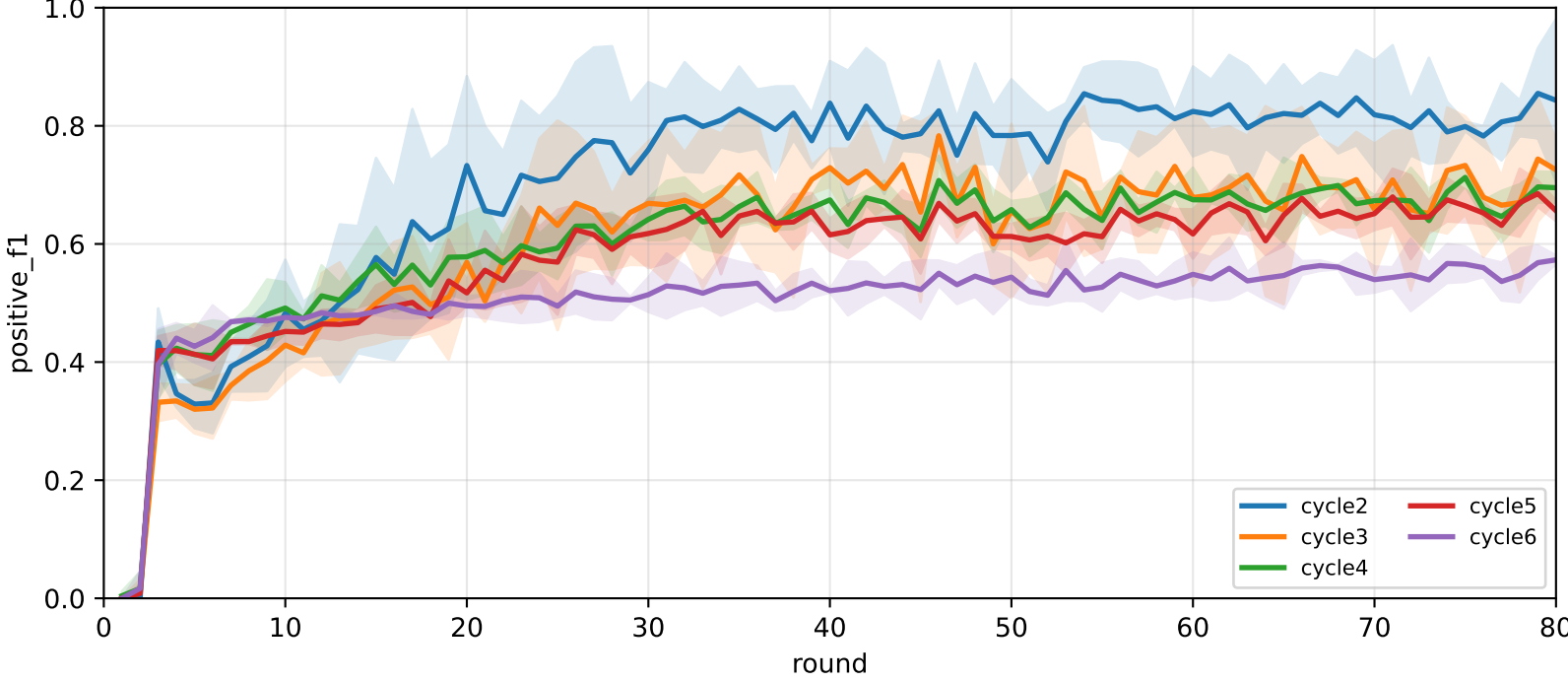
Pooled FedAvg positive F1



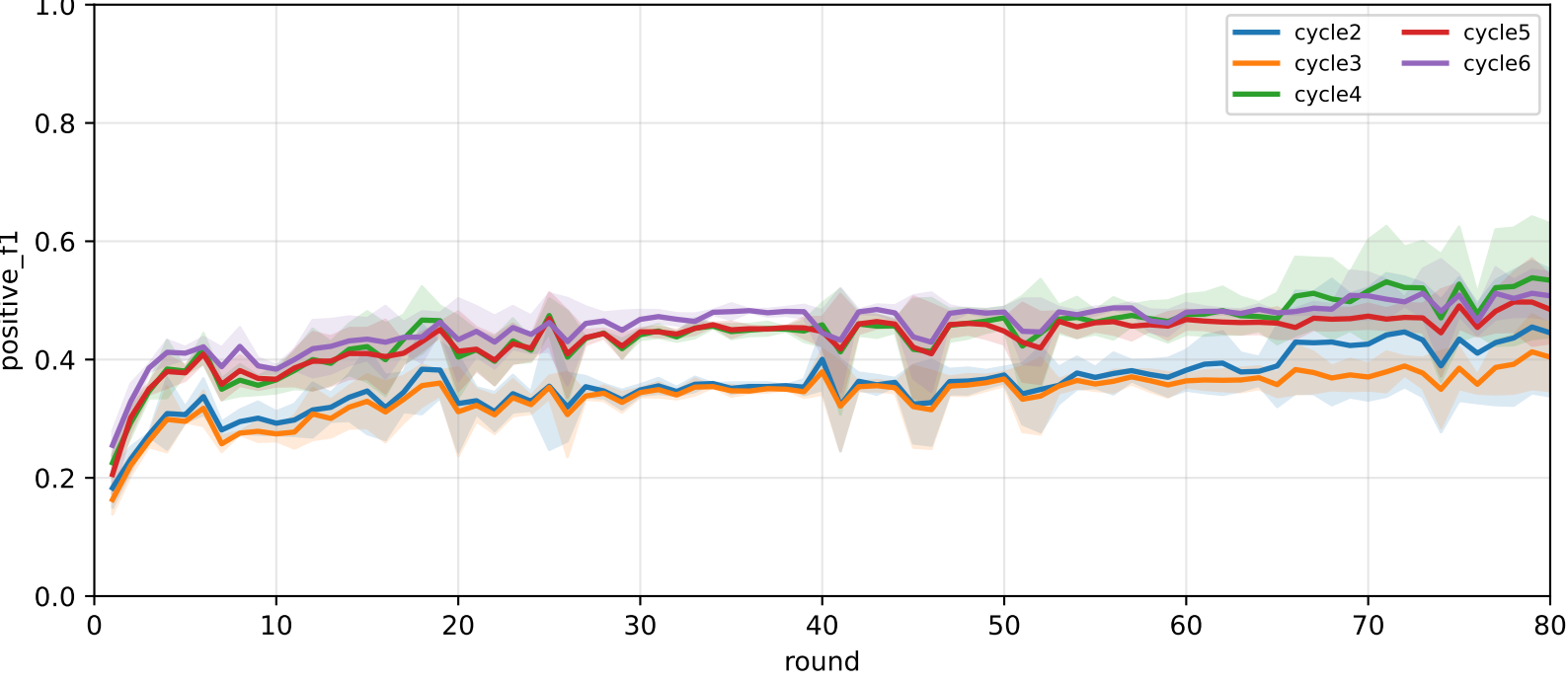
Pooled FedProx positive F1



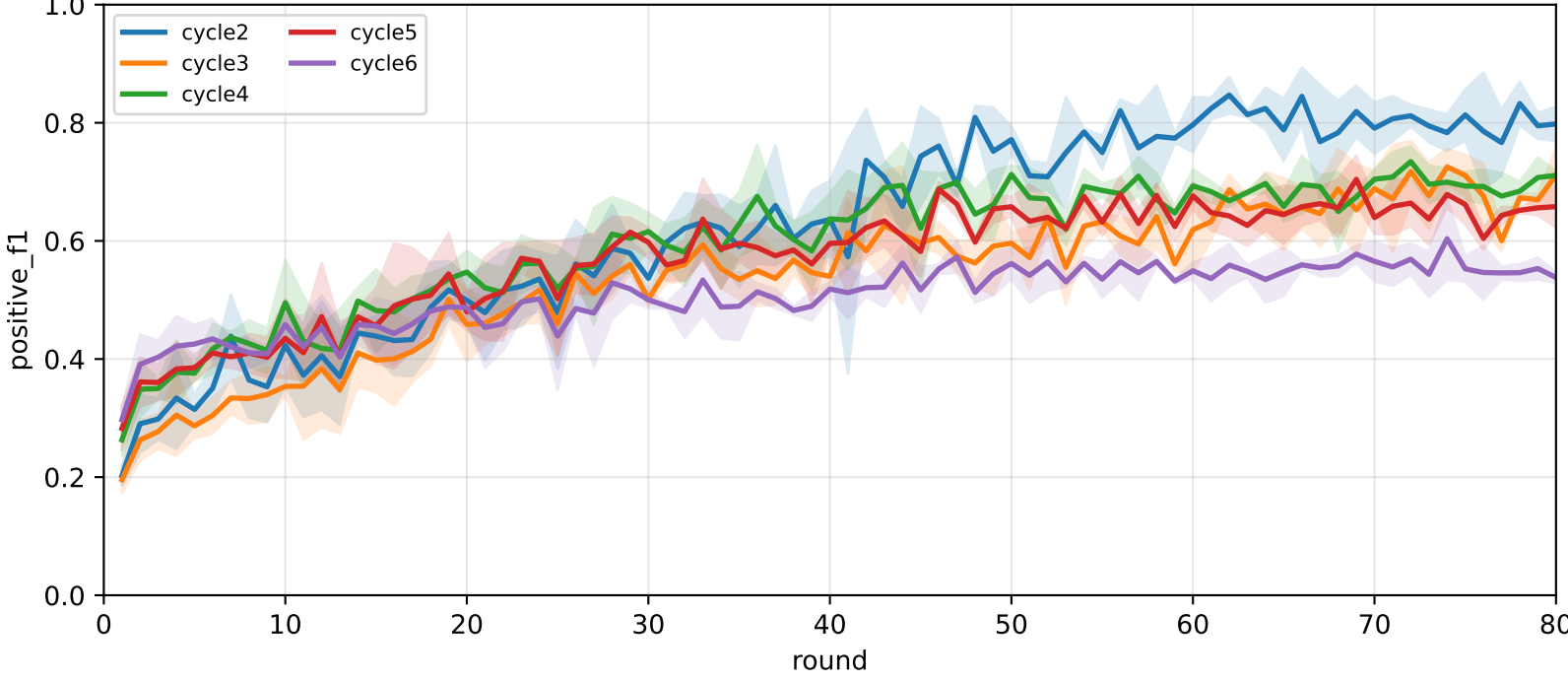
Pooled GCFL+ positive F1



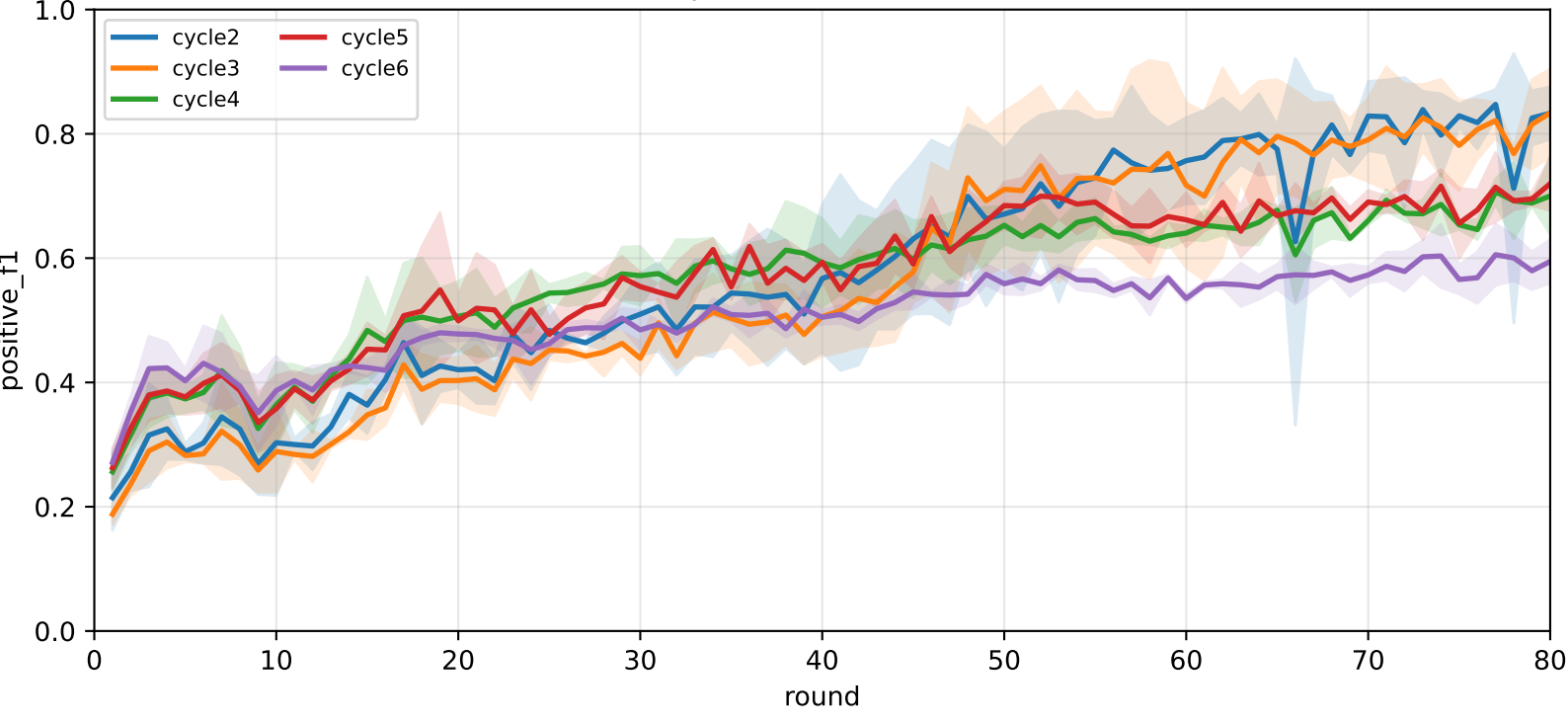
Pooled APPLE positive F1 (after-local selected model, rounds=80)



Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)

